

# **SansEC Wireless Strain Sensor for Softgood Habitats and Inflatable Space Structures**

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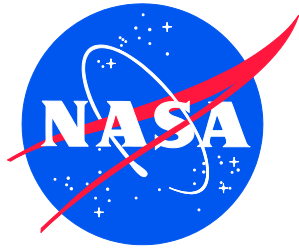
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## Nomenclature

### Acronyms

|        |                                                  |
|--------|--------------------------------------------------|
| AI     | Artificial Intelligence                          |
| ASIC   | Application-Specific Integrated Circuits         |
| CNN    | Convolutional Neural Network                     |
| DIC    | Digital Image Correlation                        |
| EM     | Electromagnetic                                  |
| F/m    | Farad per Meter                                  |
| GHz    | Gigahertz                                        |
| GSI    | Gyra Systems, Inc.                               |
| IoT    | Internet of Things                               |
| kHz    | Kilohertz                                        |
| MAE    | Mean Absolute Error                              |
| MHz    | Megahertz                                        |
| MISSE  | Materials International Space Station Experiment |
| ML     | Machine Learning                                 |
| NA     | North America                                    |
| NN     | Neural Network                                   |
| PCA    | Principal Component Analysis                     |
| RF     | Radio Frequency                                  |
| RMSE   | Root Mean Square Error                           |
| SansEC | Sans Electrical Connectors                       |
| STI    | Sierra Turbines, Inc.                            |
| VNA    | Vector Network Analyzer                          |

### Symbols

|                |                                                     |
|----------------|-----------------------------------------------------|
| $R^2$          | R-squared                                           |
| $S_{11}$       | S-parameters magnitude and phase                    |
| $X_s$          | series reactance                                    |
| $Z$            | impedance                                           |
| $\mu\epsilon$  | microstrain                                         |
| $\epsilon_r'$  | real part of the complex relative permittivity      |
| $\epsilon_r''$ | imaginary part of the complex relative permittivity |

## Abstract

*A primary concern for astronauts living in inflatable softgood structures during long duration missions is the restraint strap's steady creep while under load which could lead to catastrophic failure. To ensure astronaut safety, a mitigation method is needed to monitor the strap's strain to know when safety margins are being breached. This report summarizes an investigation to use SansEC sensor data incorporated with machine learning to predict the strap's strain while under load conditions to provide diagnostic and prognostic monitoring of the strap's integrity. SansEC sensor measurements were collected on a 12,500-pound rated Vectran strap to infer axial strain while it was loaded repeatedly between 5000- and 8000-pound loads. DIC measurements were simultaneously collected to provide true strain values to coincide with the SansEC response. Three industry partners (ResCon Technologies, VOLT AI, and Sierra Turbines) participated in this study to develop machine learning models to predict axial strain based on the SansEC measurements. All three partners successfully modeled the strap behavior and achieved an  $R^2$  above 0.97. This study has shown that SansEC sensors are able to measure strain on the Vectran strap and should be considered a candidate sensor to provide structural health diagnostic and prognostic monitoring for inflatable habitats.*

## 1.0 Introduction

NASA is developing softgood inflatable habitation modules for astronaut housing as a means to reduce launch volume. These structures are composed of several constituent layers including an inner liner, bladder, structural restraint layer straps, micrometeoroid/orbital debris protection system and a thermal control/atomic oxygen protective layer [1]. One of the primary concerns for long duration missions is material failure of the inflatable softgood's restraint straps due to creep. To ensure astronaut safety, strap strain monitoring will be essential. NASA's patented Sans Electrical Connectors (SansEC) sensor technology is one candidate approach being considered to provide diagnostic and prognostic monitoring of the strap's integrity. Its potential implementation will require less wiring than other sensor networks and reduce overall operational risks in the packaging, storage and deployment of the inflatable structure. SansEC's inventor, Dr. Stanley Woodard, previously demonstrated SansEC's capability to detect micro meteoroid orbital damage on softgoods at the White Sands Test Facility on the Hypervelocity test stand [2].

SansEC sensors are passive, wireless, open circuit electromagnetic (EM) resonators. The fundamental principle enabling SansEC sensing operations is Faraday's Law of Induction which states that a voltage is induced on a conductor when exposed to a time varying magnetic field. SansEC is unique in that it operates in the frequency domain (Analog) versus time domain (Digital) with regards to how a phenomenon is quantified. SansEC sensors are very simplistic devices composed of a continuous conductive trace looped around in a spiral configuration [3]. The trace conductivity establishes the sensor resistance while the gap spacing between adjacent trace loops establishes the sensor's capacitance and inductance, which by definition is an EM harmonic resonator. The mechanical analogy of a SansEC EM resonator is a ringing bell. When a bell is struck, it rings at its fundamental audio frequency. Immersing the ringing bell in water results in a



change in the audio frequency. SansEC sensors are “rung” and wirelessly powered from external oscillating magnetic fields typically coupled to the sensor via a near field loop antenna. Near-field coupling refers to electromagnetic fields that exist very close to a transmitting antenna or other Radio Frequency (RF) source.

A vector network analyzer (VNA) connected to the loop antenna is used both as a transmitter to provide the excitation to the sensor, and as a receiver to measure its resonance response. The amplitude and phase of the return signal received from the loop antenna, (denoted by S-parameters S11 amplitude and phase) are measured by the VNA. S-parameters are a fundamental tool in RF and microwave engineering used to describe the behavior of electrical networks and components. They quantify how much of a signal is reflected back, transmitted through, or transferred between the ports of a network. Most modern VNAs offer numerous parameters derived from the S11 response. From a machine learning perspective, these additional parameters can offer significant opportunities for distinctive features that may be beneficial to algorithmic generalization. Examples of these derived parameters are series resistance ( $R_s$ ), series reactance ( $X_s$ ), and impedance ( $Z$ ). The free space coupling distance between the sensor and near field antenna enables applications where electrical wiring is not practical to implement or will degrade the accuracy of a measurement. Once the sensor is excited by the external magnetic fields, the sensor reradiates an EM signal with a unique resonance response dependent upon the local environment’s electrical impedance. Changes in the environmental impedance adjacent to the sensor are reflected in the sensor’s resonance response. The response is picked up by the loop antenna and recorded by the VNA, which enables the sensor to monitor for changes in its immediate sensing region wirelessly, without batteries and without physical contact to the sensing region. This makes SansEC a unique sensor for nondestructive evaluation applications and ideally suited for implementation into softgood inflatable habitats.

The purpose of this investigation was to determine if the localized change in the dielectric values measured in the Vectran strap could be correlated to its strain using SansEC sensors. To accomplish this, SansEC sensors were placed in close proximity to the Vectran strap and measurements were collected with the strap under various loaded strain conditions. Digital Image Correlation (DIC) was used to quantify the strap strain for truth comparison. Relative changes in the strap’s dielectric value were also monitored using a capacitance test set to help inform observed behaviors in the sensor measurements. The SansEC data was processed in machine learning algorithms developed by three industry partners participating in the study through NASA Space Act Agreements. The testing was conducted in the NASA Langley 50 KIP load cell test stand #6 in the summer of 2024.

To establish industry participation in this investigation, three industry partners were awarded non-reimbursable Space Act Agreements based on their experience and unique approach in applying machine learning (ML) methodology to SansEC data sets. The three industry partners worked independently to develop their ML models to predict strain from the SansEC measurements. Below is a description of the companies who supported NASA in this research.

- ResCon Technologies, LLC is a software startup founded in 2020 as a spinout of The Ohio State University’s Technology Commercialization Office. It provides ultra-efficient machine learning solutions to clients in Aerospace, Internet of Things (IoT), and Wearables verticals. BareML™, ResCon’s proprietary software toolkit, empowers engineers and product designers to embed artificial intelligence (AI) anywhere, creating models that shine

the brightest when speed and efficiency are paramount and training data is limited. BareML grew out of ResCon cofounder Dr. Daniel Gauthier's work in the Physics-Informed and Quantum Computing spaces. His goal was to create low-data, ultra-fast ML that could run on conventional embedded hardware at the speeds required for Quantum Information Systems. BareML is the ideal software complement to next-gen IoT and wearable hardware. It enables the fastest and lowest-power ML deployments on the simplest hardware, as well as "bare metal" deployments on custom Application-Specific Integrated Circuits (ASIC).

- VOLT AI, LLC is an artificial intelligence company founded in 2023 in North Carolina. The company's flagship platform, VOLT SkyBridge, delivers a rapid, end-to-end machine learning and IoT VNA solution for RF applications – enabling users to collect, train, analyze, and deploy models in under 20 minutes. VOLT AI also leads innovation in textile-based sensing through its proprietary VOLT Smart Yarns: conductive, fabric-integrated sensors embedded with fine insulated wires. Developed by Supreme Corporation, a trusted industry leader with over 61 years of experience, and powered by VOLT AI's advanced artificial intelligence, these yarns enable consistent power transmission and precise RF data capture across textile-embedded systems.
- California based Sierra Turbines, Inc. (STI) has fostered the design, manufacture and integration of sensors with advanced machine learning algorithms of SansEC resonators, transforming these devices into versatile software-defined sensors as differentiators for its products in the energy and aerospace domains. In addition they have developed the end-to-end platform to deploy SansEC sensors into such environments. This research and development effort has benefitted significantly from teaming with Plyobotics (veteran-owned small business), run by former SansEC licensee James Phelps (former CEO of Gyra Systems, Inc. (GSI)), drawing on his deep expertise in RF/microwave sensing and related Neural Network (NN) work. James spearheaded key algorithmic advances and SansEC system designs at GSI and collaborated with Supreme Corporation, Advanced Materials Innovation, and Sierra Turbines, Inc. Through these partnerships, the SansEC ML platform has been field-tested in challenging environments ranging from pharmaceutical delivery lines to aircraft monitoring instrumentation. The team's experience with SansEC-based sensing has demonstrated exceptional versatility across multiple domains. In pharmaceutical applications, they have used machine learning models to detect subtle dielectric constant shifts associated with varying compound concentrations and formulation stages, enhancing quality control during storage and delivery. The team has further extended SansEC capabilities to monitor chemical composition dynamics in complex fluids such as aircraft fuel and oil by classifying spectral signature variations induced by viscosity and contaminant level changes for military tactical aircraft.

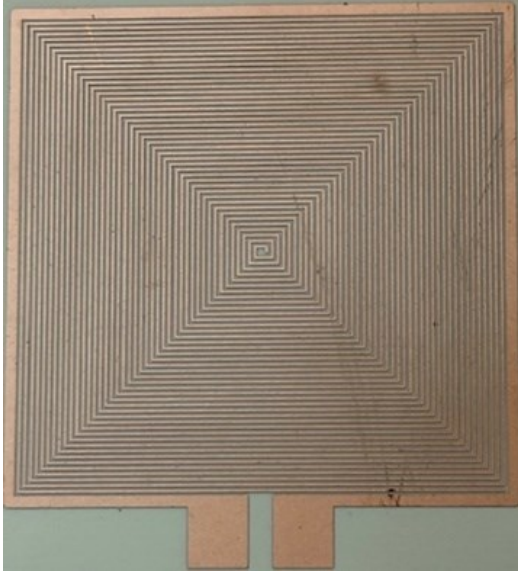
## 2.0 Approach

This investigation was conducted in two phases. The first phase focused on establishing the operational procedures to conduct repeatable and accurate measurements. This included ensuring the instrumentation were set up appropriately to capture and record the necessary data parameters to support conclusive data analysis. Initial testing indicated a potential risk the sensors could move relative to the strap during testing, so a plexiglass panel was installed in front of the strap to better secure the sensors for future tests. The information gathered in the first phase indicated the strap's material characteristic changed significantly as it was cycled repeatedly under load and established confidence in the SansEC sensor data being able to monitor and capture these changes. Once the test techniques were sufficiently improved to eliminate known uncertainties, phase two testing was initiated to collect the data used in the machine learning.

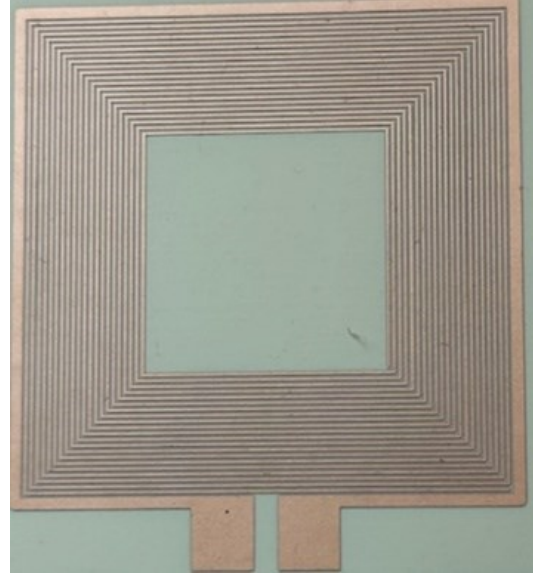
The SansEC sensors used in this study included NASA sensors that were used in previous research, and sensors fabricated by Sierra Turbines and VOLT AI. The NASA sensors were fabricated from etched printed circuit board manufacturing using half ounce copper on a 5-mil fiberglass substrate. The loop antenna around the sensor was manufactured as part of the etching process to permanently maintain its position relative to the sensor. The NASA sensors tested varied in size from 1 inch square to 3 ½ inch square to gain insight into any frequency dependencies in the measurements. The sensing range beyond the sensor surface is directly related to its resonance frequency. The lower the frequency, the greater the detection range is from the surface of the sensor. The 3 ½-inch NASA sensor was significantly larger compared to the width of the Vectran strap, but provided measurements at the upper kilohertz (kHz) frequencies.

VOLT AI, in cooperation with Supreme Corporation, provided 1-inch circular sensors manufactured from conductive yarn and stitched into a fabric backing. These sensors were of interest for expanding the design envelop in how SansEC technology could be implemented in softgood habitats. A circular loop antenna slightly larger than the sensor diameter was fabricated using semi-ridged coaxial cable to excite the VOLT AI sensor. These sensors operated in the 200 to 500 megahertz (MHz) range. The sensors performed well, and this manufacturing approach is feasible for certain implementations.

Sierra Turbines, in cooperation with Panasonic NA (North America), provided 1-inch square sensors that were manufactured on flexible electronic substrates that have been space qualified aboard SpaceX CRS-27 in 2023 aboard the MISSE Module (Materials International Space Station Experiment). The substrates are specifically designed to be used in wearable electronics and can provide another avenue to implementation. These sensors also had the loop antenna printed around the sensor, similar to the NASA sensors. These were the sensors used in the phase two data collection utilized in the machine learning algorithms in this report. Figures 2.a through 2.d present photographs representative of sensors used in this study.



**Figure 2.a Sierra Turbines 1" Square**



**Figure 2.b Sierra Turbines 1" x 5" Square**



**Figure 2.c NASA 1" Square**



**Figure 2.d VOLT AI 1" Circular**

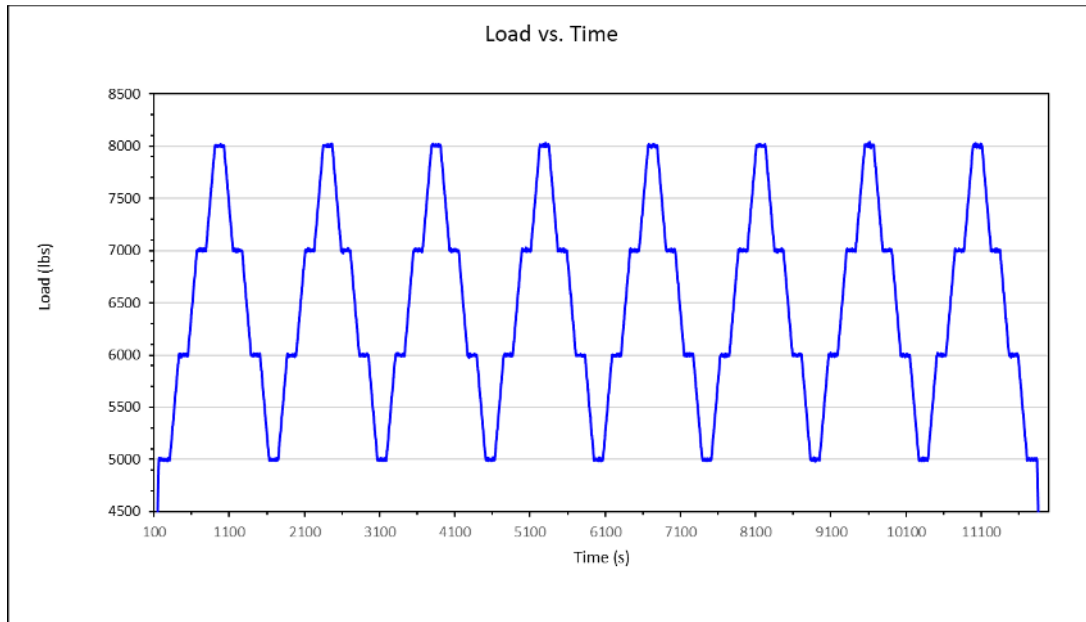
An optimal sensor configuration would operate at resonance frequencies that provide sufficient sensing depth to penetrate beyond the strap, be sensitive to changes in the strap's electrical impedance, and be appropriately sized to concentrate its coupled fields primarily to the strap. All the sensors tested during phase one had multiple harmonic resonances and sufficient signal-to-noise ratios to make trustworthy measurements. The complexities of the strap characteristics as it underwent hysteresis were not initially understood and clouded any quantifiable analysis until all the test uncertainties could be addressed. Because of this, phase one testing did not establish any obvious correlations to establish which sensors or frequency regions might outperform the others.

### 3.0 Experimental Setup

The two 1-inch wide Vectran straps used in this study were rated to 12,500 pounds. The first strap was used repeatedly to gain observational knowledge in the test operations and to optimize the data collection procedures during phase one. Eventually, fraying was observed in the stitching and it was replaced with a pristine strap to begin phase two testing. The machine learning results presented in this report are based on measurements taken on the second strap while it was still in pristine condition. For each load test, the strap was first preconditioned by rapidly loading and unloading the strap up to 8000 pounds through 5 cycles. This removed most of the excess give in the strap prior to actual strain measurements. Once this was completed, the data collection instrumentation began continuously recording test data until the conclusion of the test. The load test cell recorded the load value at a 2-second interval.

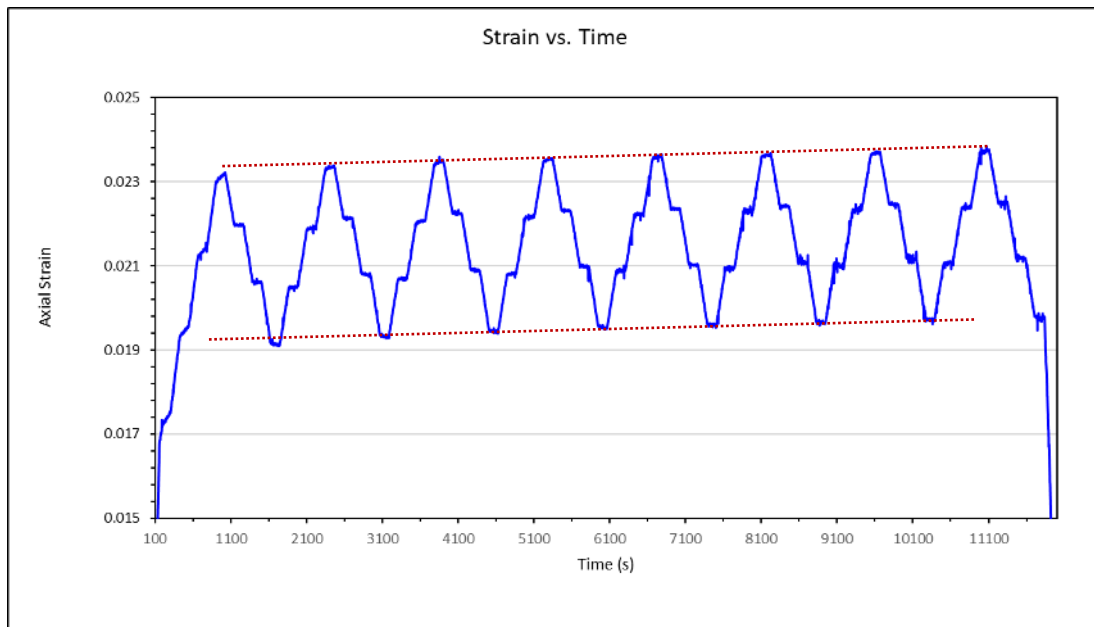
Strain measurements were collected using a digital image correlation (DIC) camera system. DIC is a non-contact optical measurement technique that determines displacement and strain on a surface by analyzing photographic images. DIC uses algorithms to track the movement of specific points (targets) on a surface across a series of images, allowing for the calculation of deformation and strain. A 4-inch section at the top of the strap was painted with a speckled pattern to facilitate the DIC measurements. The DIC data was post-processed and time synchronized to establish the axial strain at each VNA data collection. This method of strain measurement assumes the strain localized at the strap's 4 -inch painted section is also the strain seen throughout the length of the strap. This assumption has not been validated by the authors and remains an unanswered question.

The data plots shown in Figures 3.a through 3.c are from the September 10, 2024 data collection. This is the test series data that was provided to the industry partners to ingest in their ML algorithms and reported later in this report. Figure 3.a presents the load (y-axis) versus time (x-axis) plot. The plot shows that the maximum load of 8000 pounds was reached on 8 load cycles. A load cycle is defined when the load reaches 5000 pounds, increases to 8000 pounds, and then reduces back to 5000 pounds. The load value held steady for several minutes at each 1000-pound increment (5000-, 6000-, 7000-, and 8000-pound loads).



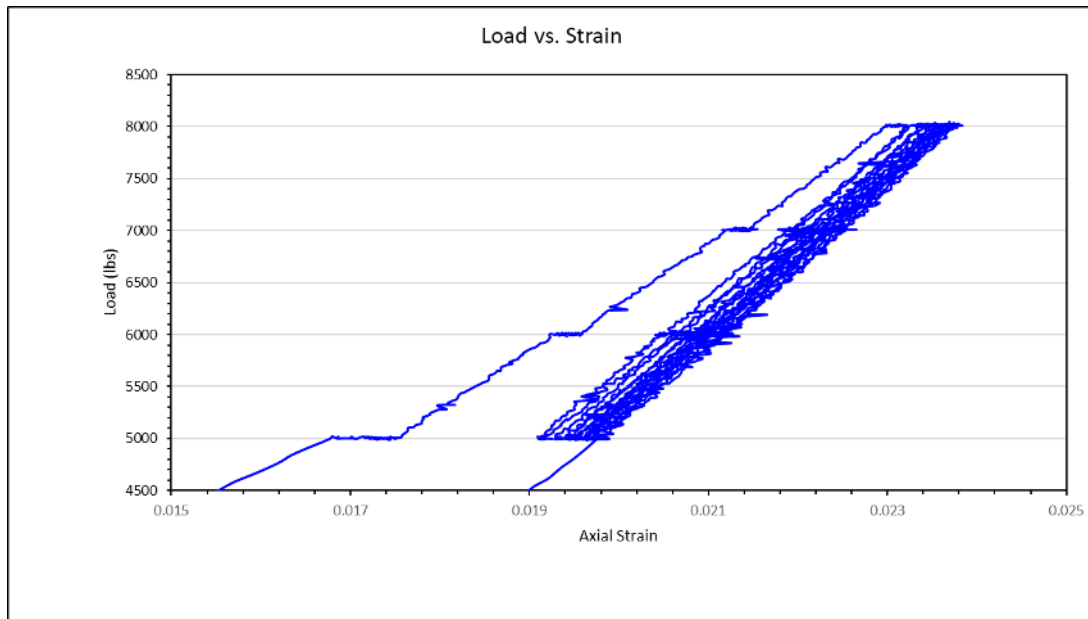
**Figure 3.a Load versus Time**

Figure 3.b presents the measured strain value from the DIC system over the 8 load cycles. The plot presents the strain (y-axis) versus time (x-axis). From this plot, it is observed the strain level continually increases with each consecutive load cycle and the strain value increases when held at a constant load value.



**Figure 3.b Strain versus Time**

Figure 3.c presents the corresponding load (y-axis) versus strain (x-axis) plot. The plot shows the first load up cycle has the most change in the strain. As the excess give in the strap is removed, the load strain curve appears to have the same slope. A close examination of the plot also shows the increasing strain when the load is held constant.



**Figure 3.c Load versus Strain**

### **3.1 Dielectric Measurement Setup**

To help characterize the nature of the dielectric changes occurring in the strap while under load, a dielectric test fixture was instrumented to collect the relative dielectric change in the strap during the initial stages of this study. An Omicron Electronics Dielectric Analyzer Spectano 100 unit was used with a Keysight Technologies, Inc. Dielectric Test Stand 16451B to record dielectric parameters while the load was held stationary at 1000-pound load increments. Due to the equipment limitations, data was only collected between 10 and 30 MHz. The strap was placed between the Dielectric Test Stand's 2-inch electrodes. Once the strap was in position, the electrodes were brought together to minimize the air gap spacing to the strap while not hampering the strap's vertical motion while under load. The data collection for this unit took 250 seconds. The strap was held at a constant load at each 1000-pound load increment while the dielectric measurements were collected. Figure 3.1.a shows a photograph of the dielectric test stand near the bottom of the strap as installed on the load test stand during the phase one test series. The photograph also shows the speckled painted portion at the top of the strap used for DIC strain measurements. In the middle of the strap are a VOLT AI 1-inch circular conductive yarn sensor (uppermost), a NASA 1-inch square sensor (middle) and a NASA 3 ½-inch sensor (bottom).





**Figure 3.1.a Phase One Test Setup with Dielectric Test Stand**

### **3.2 SansEC Measurement Setup**

The instrumentation used to collect the SansEC measurements included both a Sark 110 VNA and a Copper Mountain R140 reflectometer. The Sark's frequency range is 100 kHz to 210 MHz. The Copper Mountain unit operates from 85 MHz to 14 gigahertz (GHz). Both VNA instruments collected data simultaneously from different sensors, but their operational frequency bands never overlapped. The VNAs were connected to flexible coaxial cables holding the loop antennas which excite the SansEC sensor. The loop antenna and sensor were aligned with the strap and taped in position. To remove uncertainty in the measurement during the phase two testing, the sensors were taped to a plexiglass panel which was rigidly mounted to the load test stand. The plexiglass was positioned parallel to the installed strap and offset just enough to prevent the strap from touching the plexiglass during the tests. Figure 3.2.a shows a photograph of the SansEC sensors taped to the plexiglass panel as tested on September 10, 2024 for the phase two testing. This was the test setup for the data collection used by the machine learning algorithms. The sensors shown on the plexiglass are 1-inch square sensors on the Panasonic substrate provided by Sierra Turbines. The top sensor is a single sensor, and the bottom is a 3-sensor array. The ML results shown in this report used data from the 3-sensor array measurements for analysis.



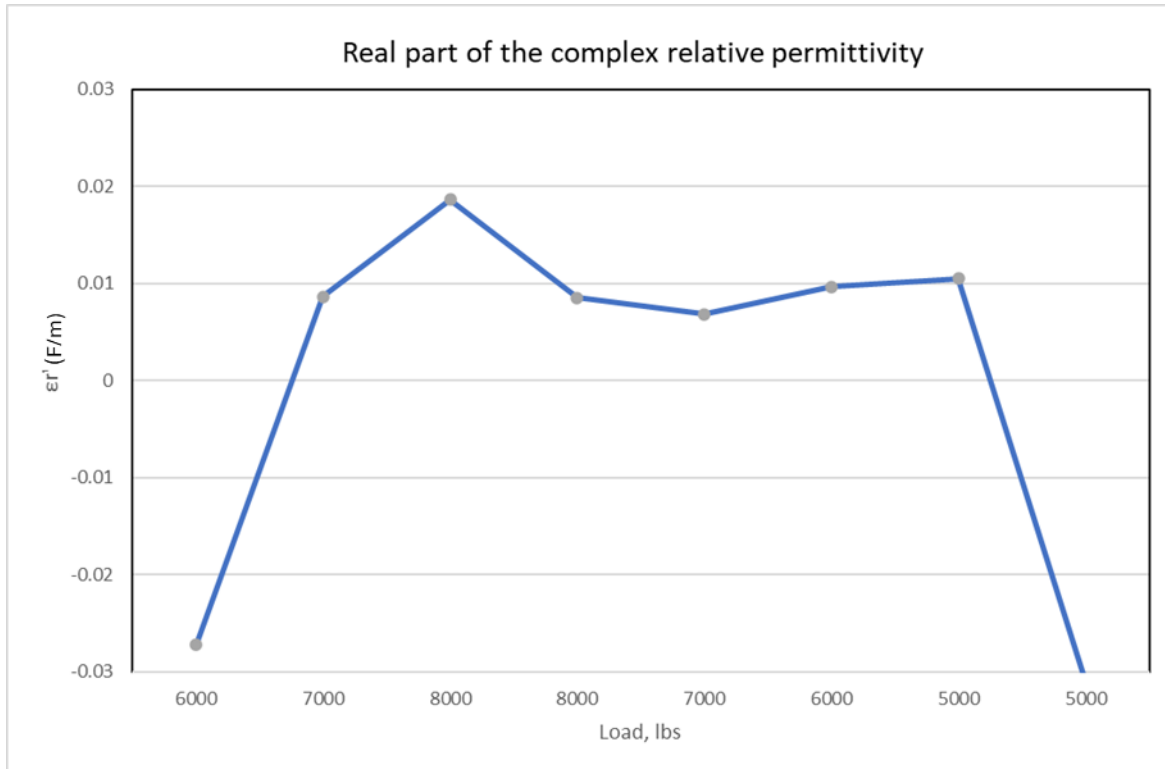


Figure 3.2.a SansEC Sensors used on September 10, 2024 Phase Two Test

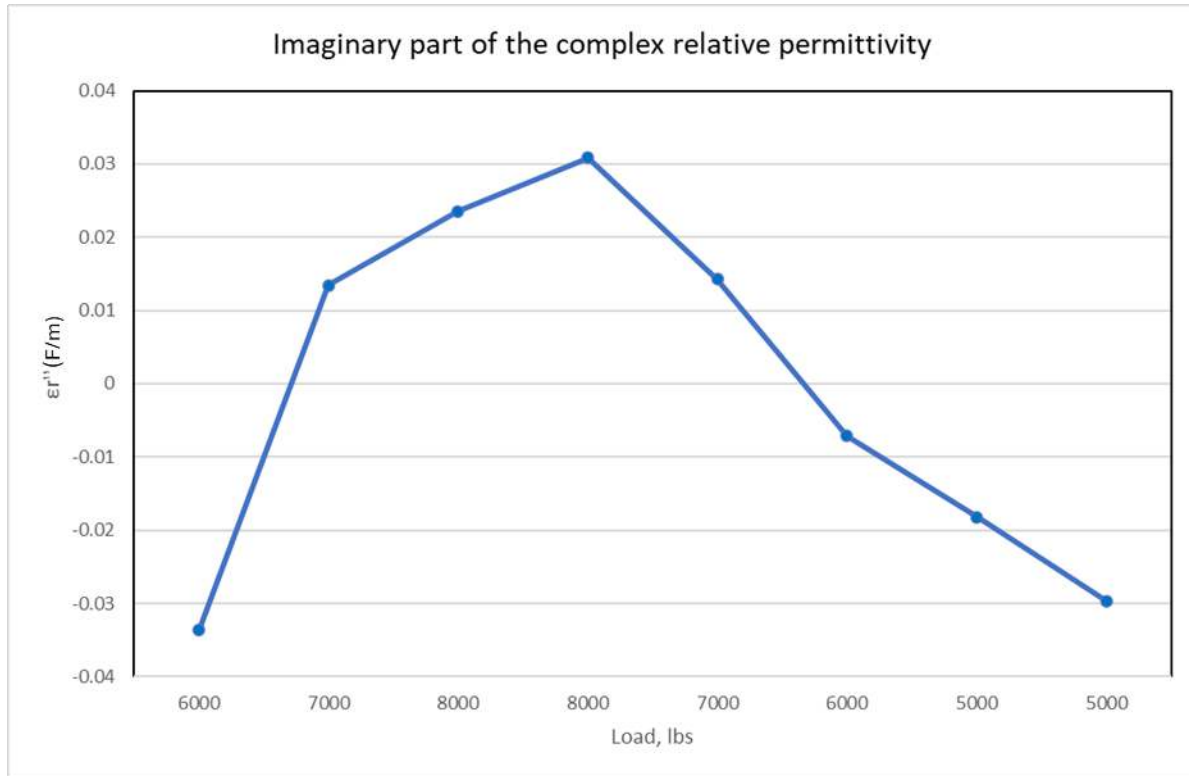
## 4.0 Dielectric Measurement Results

A dielectric test set was used to measure relative change in the real and imaginary parts of the complex relative permittivity while the strap was held at constant loads during phase one testing. The measured changes were so small they fell within the uncertainty tolerance of the measurement system and no conclusive insight was determined. Figure 4 presents an example of typical dielectric measurement data that was collected. Figure 4.a presents the relative change of the real part of the complex relative permittivity ( $\epsilon_r'$ ) at 25 MHz for one load cycle from a test conducted on July 15, 2024. The data plots show the relative change in the measured permittivity compared to the initial 5000-pound load measurement. The x-axis label indexing indicates the strap load when the data was collected. Two collections were taken back-to-back at 8000 and 5000 pound loads. Figure 4.b presents the corresponding data for relative change of the imaginary part of the complex relative permittivity ( $\epsilon_r''$ ).

Absolute relative permittivity measurements were also conducted on an unloaded 12,500 pound rated Vectran strap and the permittivity was in the range of 2.75 Farads per meter (F/m) for frequencies from 200 MHz up to 20 GHz.



**Figure 4.a Relative Change in  $\epsilon_r'$  at 25 MHz**



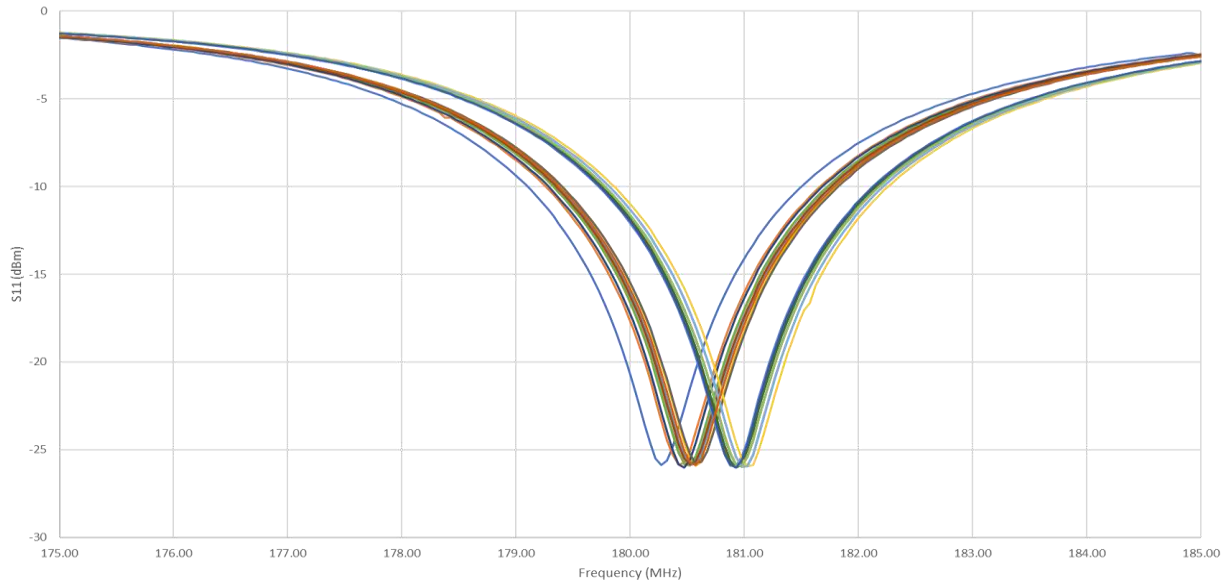
**Figure 4.b Relative Change in  $\epsilon_r''$  at 25 MHz**

## 5.0 Real Time Observation of Strap Degradation

The SansEC sensors picked up an insightful observation into the strap's strain dynamics during the sixth load test series using the same 12,500-pound rated Vectran strap during phase one testing. The data collection occurred on August 28, 2024, from two SansEC VNA instruments measuring two different SansEC sensor geometries. The test comprised three full 5000-to-8000-pound load cycles. During the second 8000-pound cycle, the SansEC response changed abruptly; this change appears to be associated with observing real time strap degradation. The data showed what at first appeared to be an anomaly in the data collection until the abnormality was confirmed simultaneously in both SansEC measurement instruments. The sensors used in this test were manufactured by Sierra Turbines. The Sark 110 VNA collected data from a 1-inch square sensor between 130 and 230 MHz with a 50 kHz frequency step. The Copper Mountain R140 VNA collected data from a sensor with a 1-inch outer dimension and 0.5-inch inner dimension. This sensor was operated from 240 to 540 MHz at a 15 kHz frequency step.

Figure 5.a presents a 10 MHz wide portion of the SansEC S11 amplitude (y-axis) versus frequency (x-axis) centered at 180 MHz. The 19 traces displayed in the plot represent the 5000-, 6000-, 7000- and 8000-pound load data collections for the three load cycles. The plots are not labeled because they could not be distinguishable at this viewing scale. Three distinct patterns emerge in this data. The single outlier curve on the far left corresponds to the first 5000-pound hold data. The excess elasticity in the strap has not yet been pulled out. The next series of curves to the right are bundled tightly together and represent the first 6000-pound load through the second upward cycle 8000-pound load data. The next series of traces to the far right correspond to the second downward load cycle through the full third load cycle and show that the traces are still tightly bundled but have

shifted significantly to a higher frequency. This result was very unexpected and further analysis was needed to determine the cause. The frequency shift occurred when the strap was being held at 8000-pound load on its second cycle.



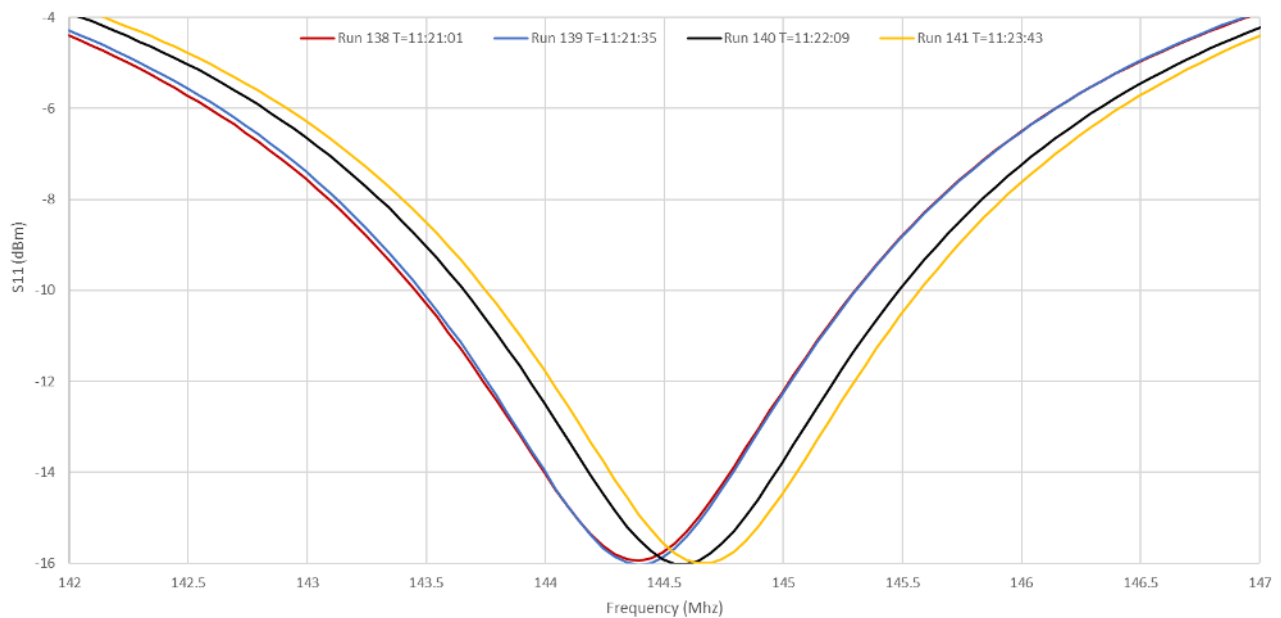
**Figure 5.a SansEC S11 Amplitude versus Frequency centered at 180 MHz**

Figures 5.b, 5.c and 5.d present the SansEC response for four sequenced data collections for the times just before and just after when the frequency shift occurred. Each figure displays a different portion of the SansEC frequency band to enable visualization. The traces are labeled by their Sark run number and time stamp when the data was recorded. The Sark VNA will sweep through its frequency band starting at 130 MHz and steps through each frequency step until it stops at 230 MHz when the data is recorded.

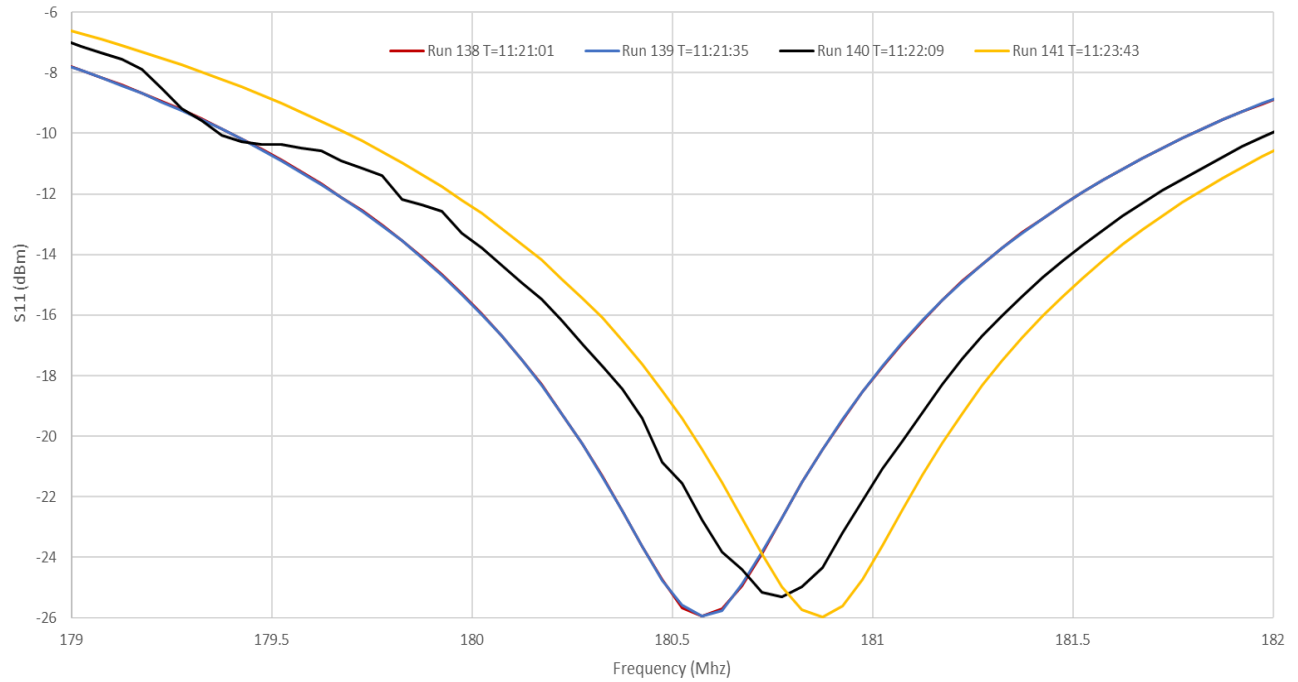
For this discussion we will evaluate the trace behavior in sequential time order of the data collection. The first time-sequenced data collection shown in the plots is run 138 (red trace), collected at 11:21:01 when the strap load was first holding at 8000 pounds. The second data collection, run 139 (blue trace), was collected 34 seconds later with the load holding steady at 8000-pounds. These two traces essentially overlay at 144 MHz and 180 MHz resonances (Figures 5.b and 5.c), but at 216 MHz (Fig 5.d), run 139 deviates significantly. This is the first indication of a change in the resonance response. The behavior of the response mimics how the response would change if the strap were moved suddenly, or an instrumentation glitch occurred.

The next data collection is run 140 (black trace) at time 11:22:09 with the load still holding constant at 8000 pounds. The shift in the resonance frequency response at 144 MHz (Fig. 5.b) is readily apparent, but otherwise the data is behaving as expected. At 180 MHz (Fig. 5.c), the signal again looks like either the strap was jolted or had a bad data collection. The depth of the resonance has changed slightly compared to all other data collections which could indicate the strap distance to the sensor may have changed. At 216 MHz (Fig. 5.d), the sensor behavior is back to normal but remains well shifted compared to the previous runs. The final trace is run 141 (yellow trace), collected at 11:23:43 just before the load begins to be reduced. The strap measurements have restabilized to their new state, and at 216 MHz (Fig 5.d), it overlays nicely with run 140.

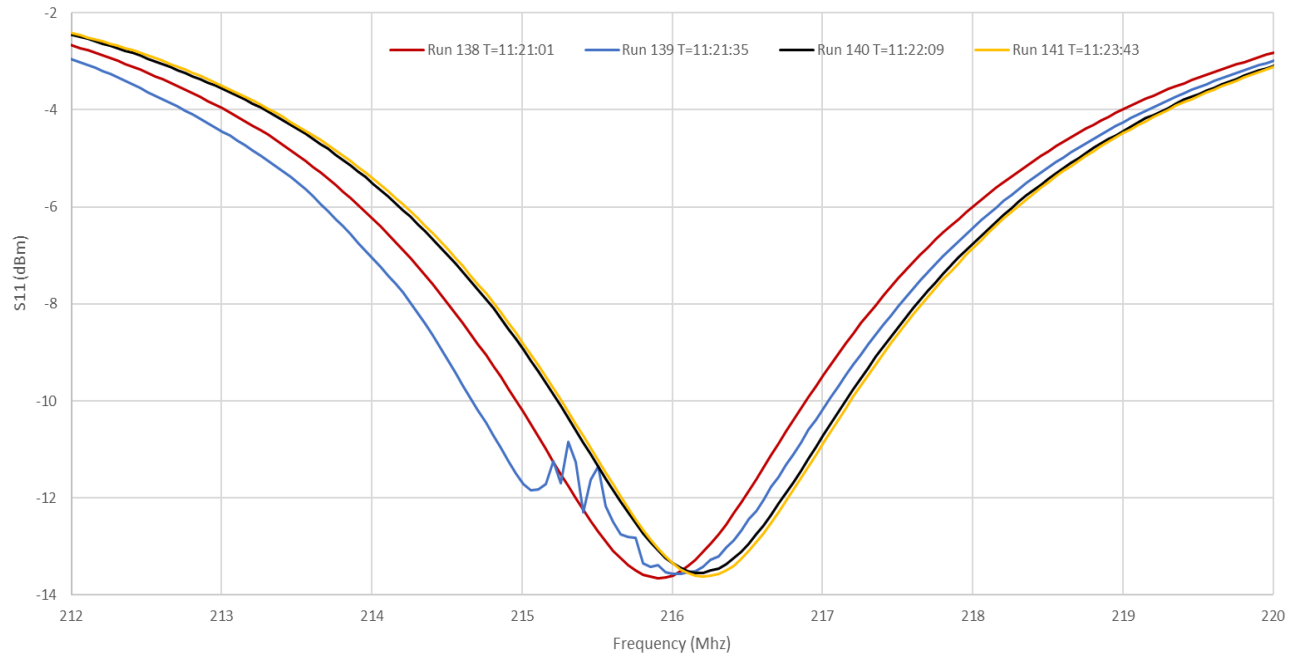
Though not presented in this report, the Copper Mountain analyzer data was examined and showed a data anomaly at this same time stamp indicating it was likely not an instrument glitch occurring in both measurements simultaneously. Examination of the strain data did not show any significant change beyond the normal noise level associated with the measurement. From this analysis, it appears that some of the threads in the strap broke at two different times causing a dynamic oscillation in the strap which was picked up in the sensor response. The thread breaks were a common occurrence during this investigation and were heard audibly as a loud ping. The weave in the strap was able to immediately redistribute the load and maintain a constant strain at the location of the DIC measurement. The localized strain implicitly measured at the SansEC sensor location can only be inferred from the DIC measured strain values and it is unknown if there could be a strain gradient along the axis of the strap which the sensor would have monitored.



**Figure 5.b SansEC S11 Amplitude versus Frequency centered at 144 MHz**



**Figure 5.c SansEC S11 Amplitude versus Frequency centered at 180 MHz**



**Figure 5.d SansEC S11 Amplitude versus Frequency centered at 216 MHz**

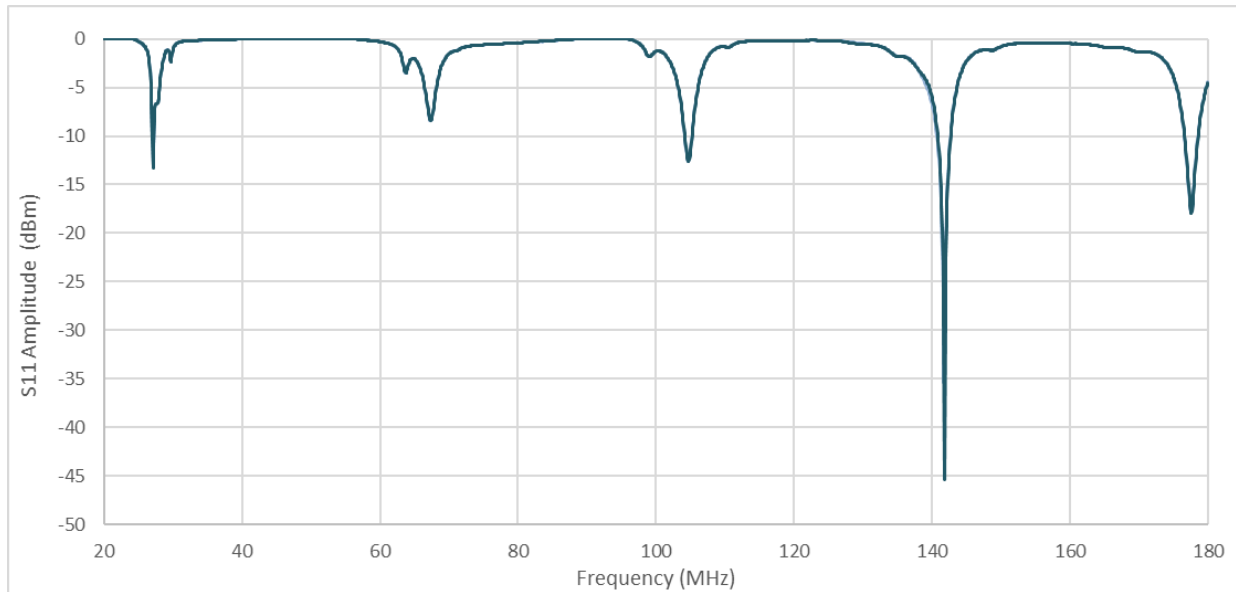
The analysis to investigate the data anomaly also led to the observation that the frequency change of a SansEC resonance alone would likely not provide a measure of the strap's strain when the strap had been exposed to variable load cycles. If we assume the DIC measured strain is homogenous across the length of the strap, and the SansEC resonance response can shift 500 kHz

at the same strain values, additional SansEC signal information would be required. Possible signal parameters such as the frequency bandwidth between multiple resonances, phase information, or a derivable electrical component parameter calculated from the electrical impedance formula may be essential features to support the ML prediction. The correlation between the strap's strain and the SansEC response is much more complex than a simple linear relationship to its resonance frequency, as will be revealed in the phase two test observation discussion summarizing the machine learning results. The strap's material characteristics are tied to the physical state of the weave binding the threads. As the strap strain increases, either from increased loads or from being held at a constant load, the individual threads achieve an equilibrium where the weave locks the threads in place, hampering the continued elongation in the strap system. If the strap is relaxed with the threads locked in the weave, the strain will remain higher at the same load than its pre-equilibrium condition. An example of this is shown in Fig 3.b in the strain versus time plot. For each consecutive load cycle, the strain gradually increases and does not reduce to its previous value. At each constant load hold position, the strain gradually increases, creating a bias in the load versus strain curve. It's interesting to note that the slope of the load versus strain curve in Fig 3.c remains constant after the initial elasticity is removed, but a bias in the strain values is seen with each consecutive load cycle.

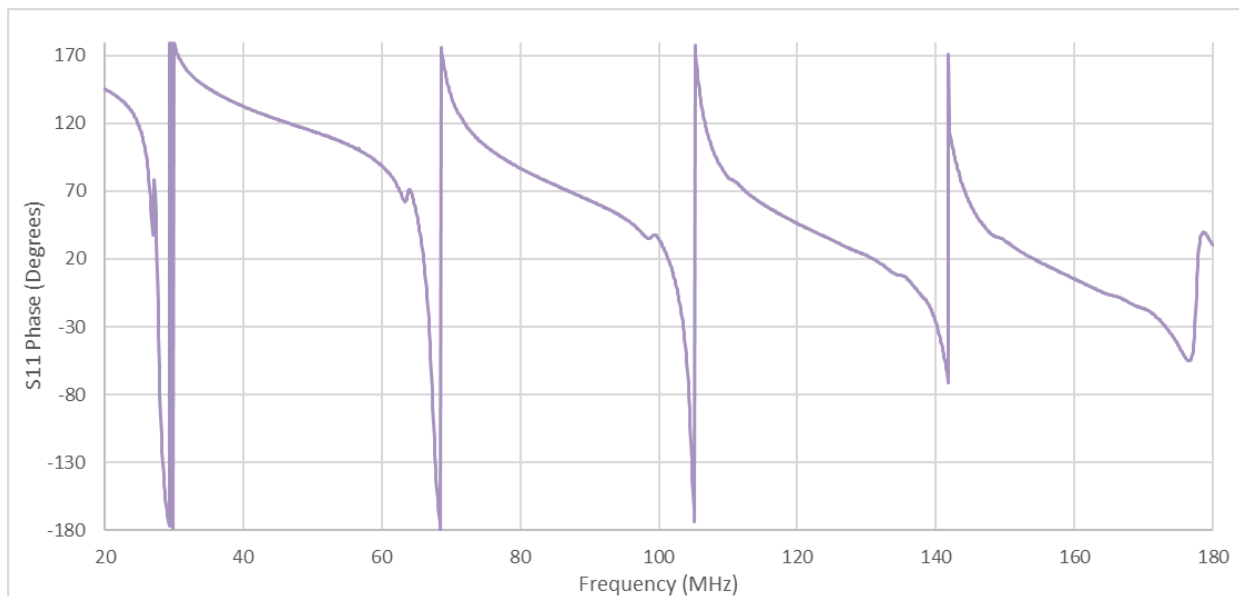
Further testing would be needed to determine if the resonance frequency change alone provides a quantifiable relative measure of strain if the quantification were only limited to when the strap is under a constant load (not cycled). The SansEC measurement response is relying on the physical state of the strap's geometry to infer strain in addition to any material property changes that might be occurring. The thickness of the strap is believed to get smaller as the threads are pulled tighter increasing the strain until its minimum thickness is reached and the strap begins to fail. Load to failure testing was outside of the scope of this experiment but is a logical next step.

## **6.0 Phase Two Test Observations**

To quantify the strap's absolute strain values based off the SansEC response, machine learning techniques were investigated by NASA's Space Act Agreement Partners, ResCon Technologies, Sierra Turbines, and VOLT AI. The load test data collected on September 10, 2024 was provided to the NASA industry partners to incorporate into their respective machine learning algorithms to estimate the absolute strain from the SansEC response. Each partner developed their own unique machine learning approach, which will be discussed separately. The plots shown previously in Figure 3.a to 3.c represent the load cycle data from this test and Figure 3.2.a shows the SansEC sensors that were used. The SansEC data ingested into the algorithms were collected using the SARK110 VNA from 20 MHz to 180 MHz with an approximately 10 kHz frequency step (1602 frequency points). Figure 6.a presents a representative SansEC S11 amplitude versus frequency plot. S11 refers to a VNA single port reflection coefficient measurement, which provides the amplitude and phase at each frequency point. Figure 6.b presents the corresponding S11 Phase plot.



**Figure 6.a S11 Amplitude versus Frequency**



**Figure 6.b S11 Phase versus Frequency**

## 7.0 ResCon Technologies Machine Learning Summary

ResCon Technologies employed their state-of-the-art “supervised learning” algorithm known as next-generation reservoir computing to develop a model of the relationship between strap strain and the SansEC signal. ResCon limited their analysis to only include the S11 amplitude parameter and did not find that the S11 phase significantly improved the prediction performance of their model. A total of 49 data files were included in the analysis, one file for each time the load was held at 5000, 6000, 7000, and 8000 pounds over the eight load cycles. To minimize training requirements, a “resonance finding” algorithm was developed to automatically identify the

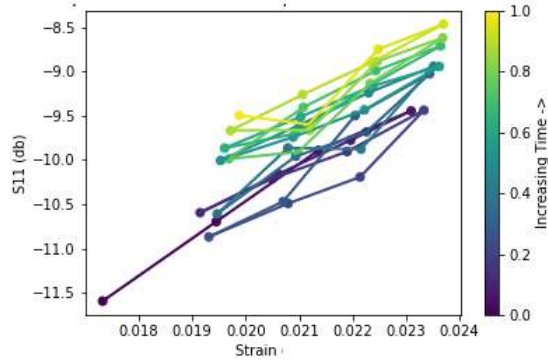


frequency points in the SansEC response with the largest variation with strain. Only a small bandwidth of frequency points around the resonance was kept to greatly reduce the total number of data points needed. For the SansEC data to be predictive in any model, the model requires a one-to-one relation between the strain and the signal. That is, for each strain value, there can only be a single value of the SansEC signal. Note that requiring a one-to-one relation does not preclude a complex, nonlinear relation, which a machine learning algorithm can learn.

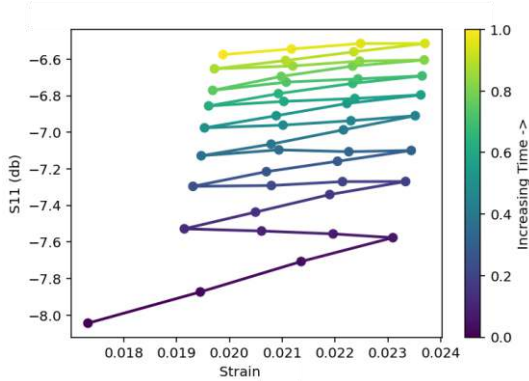
## 7.1 Data Linearity

To examine the linearity between the SansEC response and strain, ResCon analyzed the S11 amplitude at a single frequency point at the five resonances (as shown in Figure 6.a) for the 49 data files. The S11 amplitude versus strain is plotted in Figure 7.1.a through 7.1.e for the five resonance frequencies. The data is color coded to indicate which load cycle data was collected during the experiment, where dark purple indicates the first load cycle, and yellow indicates the last. During the first increasing load cycle from 5000 to 8000 pounds, the first four data points across all resonances except the one around 141 MHz appear to fall on a straight line. However, for the decreasing load part of the first cycle, we see that the next three data points fall on a straight line, but the slope is substantially lower. From these plots it is also possible to observe the increasing strain bias in each subsequent load cycle for a given load. Over the subsequent cycles, this process repeats where the slope changes on each increasing/decreasing part of the cycle. This clearly indicates that a one-to-one relation between strain and SansEC S11 amplitude response does not occur in this experiment. For all resonances except the 27 MHz resonance (Figure 7.1.a), we see that the slope of the lines during the final cycle (yellow color) is essentially equal to zero. This means that there appears to be no dependence of the SansEC response on strain based on just the S11 amplitude at a single frequency for the four highest frequency resonances. Hence, we expect that data from these resonances will give no information for predicting strain.

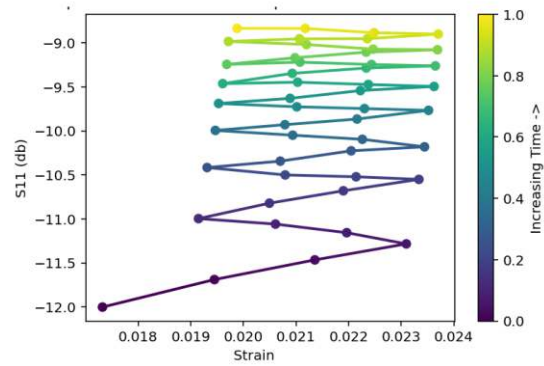
The frequency data around the upper four resonances were fed into a full machine learning analysis, which showed a large prediction error. Given this negative outcome, only the frequencies around the 27 MHz resonance were used in further analysis. The 27 MHz resonance has some promise because the slopes of the lines corresponding to each increasing/decreasing load in each cycle are roughly the same. Individual data points in each cycle have substantial variability and a single data point can vary vertically by approximately half the width of the vertical extent of the cluster of points beyond the first load value. On the other hand, the relation between the S11 amplitude and strain is not one-to-one, hence we expect some prediction error. The S11 amplitude data around the 27 MHz resonance data was down-sampled to 40 total frequency points per file. This data focusing and decimation means that the original data set of 78,000 points over all frequencies in the 49 files has been reduced to 1,960 frequency points. This reduction substantially reduces the computer time needed to train and deploy the machine learning model. The machine learning results were shown not to be sensitive to the number of datapoints, indicating that the number of data points could be further reduced.



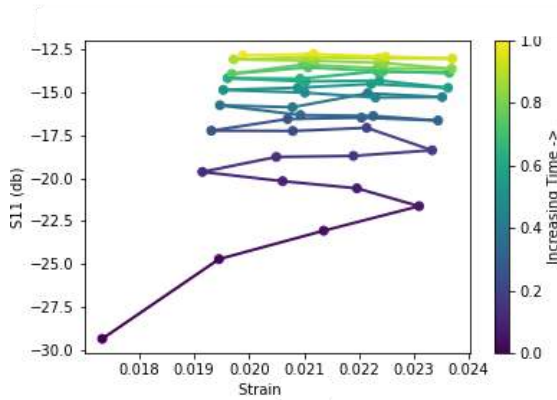
**Figure 7.1.a Amplitude versus Strain at 27 MHz**



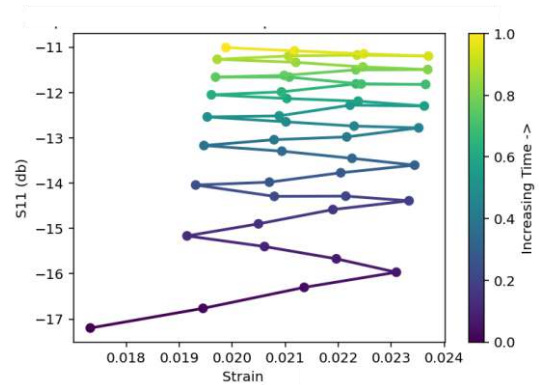
**Figure 7.1.b Amplitude versus Strain at 67 MHz**



**Figure 7.1.c Amplitude versus Strain at 104 MHz**



**Figure 7.1.d Amplitude versus Strain at 141 MHz**

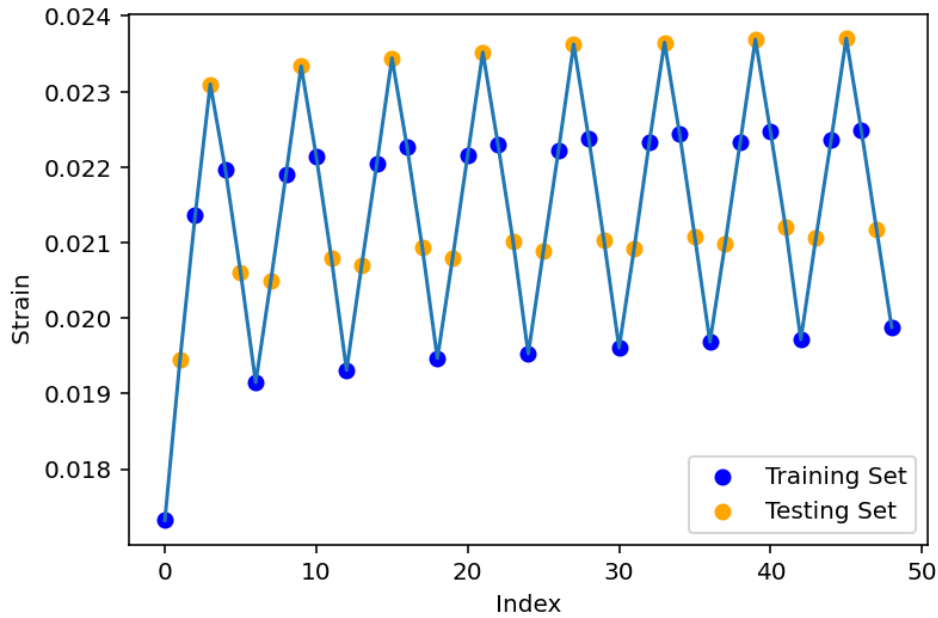


**Figure 7.1.e Amplitude versus Strain at 177 MHz**

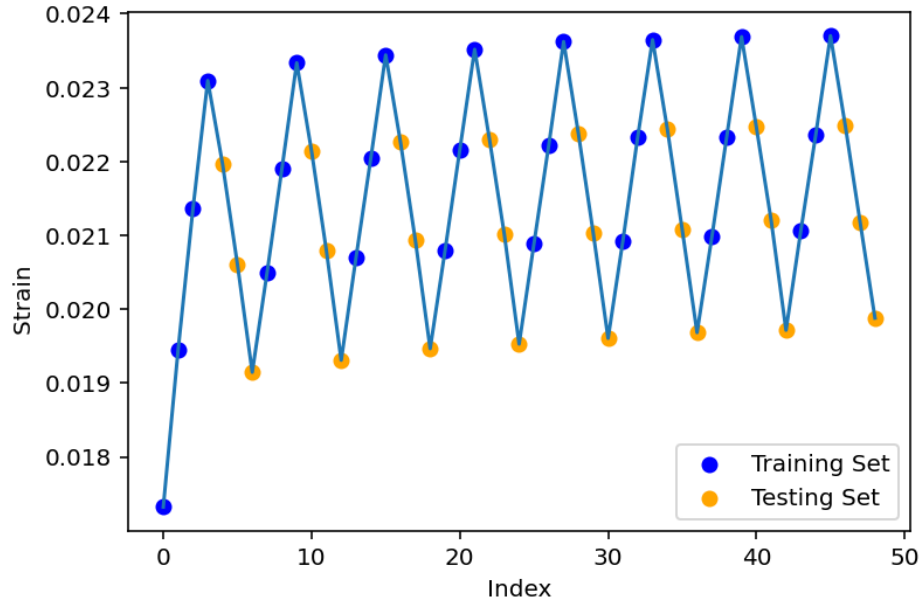
## 7.2 Training & Testing

ResCon examined several different techniques to train their algorithm. The model was first trained using the S11 amplitude data from the 40 frequency points around the 27 MHz resonance over the first few load cycles (e.g., the first cycle only, the first two cycles, the first three cycles, etc.) and then the trained model was used to predict the strain for the subsequent cycles. This approach performed poorly: the errors - normalized to the maximum change in strain over the experiment - exceed 50%. The poor performance of this training procedure is due to the fact that there is not a

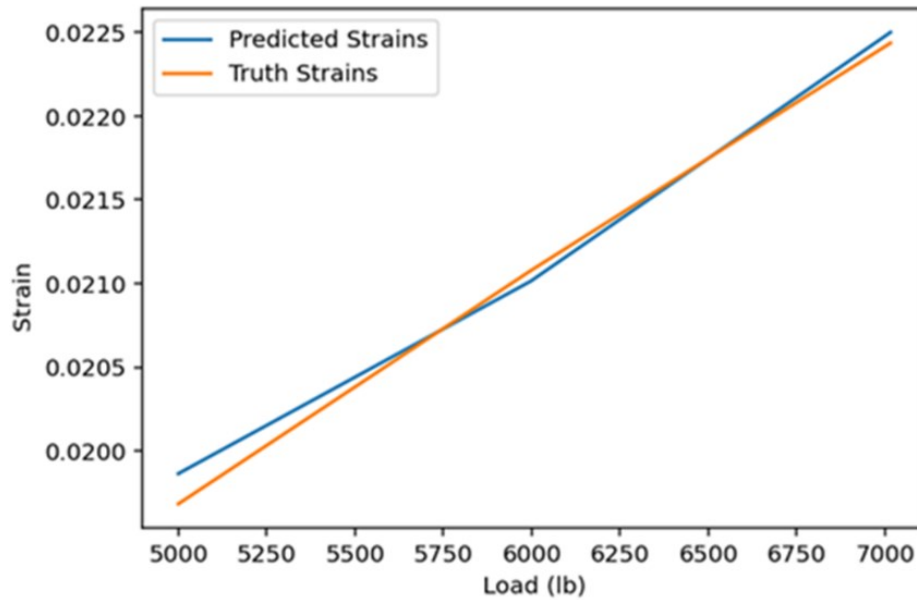
one-to-one relation between the SansEC signal and the strain as discussed in Section 5. To improve the overall model performance, two other strategies were adopted to account for the change in the S11 signal over the course of the load cycling. One method interspersed training data with the testing data, thereby creating temporally “local” machine learning models. Another method included adding a temporal feature to the machine learning model that keeps track of when the data was collected in the experiment. For each data file, an index was assigned starting at zero corresponding to the the first data file, and index 49 corresponding to the last file. Adding a temporal index feature is common in the machine learning community, especially for state-of-the-art models for natural language process (also known as large language models). Protocol 1 used every other load data file for training and the trained model was used to predict the remaining points as illustrated in Figure 7.2.a. Protocol 2 used the increasing load part of the sequence for training the model and the decreasing load files were used to predict their strain as illustrated in Figure 7.2.b. For each protocol, two experiments were performed using either just the S11 amplitude or the S11 amplitude and temporal index variable. The S11 phase did not show any improvement in the training results and was excluded from further investigations. Figure 7.2.c presents a typical comparison result between the model’s predicted strain and the measured test data strain for the sixth load cycle from Protocol 2. The slope of the strain as a function of load is similar for both sets of data. For both of these protocols, the model is trained on approximately 50% of the data and tested using the remaining data.



**Figure 7.2.a Strain versus Index for Protocol 1**



**Figure 7.2.b Strain versus Index for Protocol 2**



**Figure 7.2.c Strain versus Load Prediction for the Sixth Cycle Using Protocol 2**

The  $R^2$  (R-squared) metric is a statistical measure used to define how well the prediction from an ML regression model matches the truth data. It is also known as the coefficient of determination.  $R^2$  values range from 0 to 1 with a value of 1 indicating the model matches perfectly. It is calculated using the formula:  $R^2 = 1 - (SSR / SST)$ , where SSR is the sum of squared residuals and SST is the total sum of squares. Table 7.2 provides the  $R^2$  metric for the two protocols. The lowest error is obtained using Protocol 1, where there is always a training datapoint that neighbors a testing point. This corresponds most closely to the testing procedures used by VOLT AI and Sierra Turbines, where a small subset (20% or less) of the entire dataset was selected for testing the ML algorithm.

For Protocol 1, the lowest testing error is ~3%, corresponding to  $R^2 = 0.97$ , which demonstrates good model generalization.

**Table 7.2 Machine Learning Model Prediction Results Using Protocol 1 and Protocol 2 Training Methods Using All Eight Load Cycles.**

| <b>Machine Learning Method</b> | <b>Features</b>                | <b>Frequency Points</b> | <b>Training Samples</b> | <b>Test Samples</b> | <b>Training <math>R^2</math></b> | <b>Testing <math>R^2</math></b> |
|--------------------------------|--------------------------------|-------------------------|-------------------------|---------------------|----------------------------------|---------------------------------|
| Protocol 1                     | S11 Amplitude                  | 40                      | 25                      | 24                  | .970                             | .950                            |
| Protocol 2                     | S11 Amplitude                  | 40                      | 25                      | 24                  | .983                             | .922                            |
| Protocol 1                     | S11 Amplitude & Temporal Index | 40                      | 25                      | 24                  | .977                             | .969                            |
| Protocol 2                     | S11 Amplitude & Temporal Index | 40                      | 25                      | 24                  | .983                             | .952                            |

## 8.0 VOLT AI Machine Learning Summary

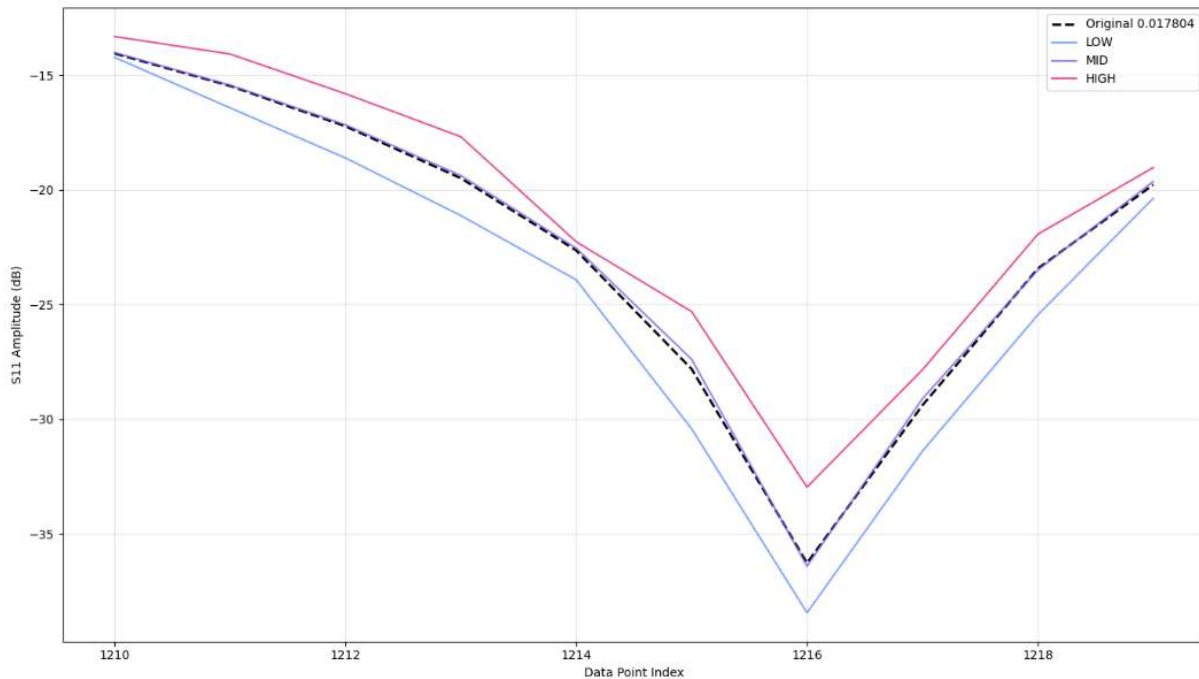
VOLT AI developed three different machine learning models to evaluate the efficacy of using the SansEC S11 amplitude and phase parameters to predict strap strain. These included a Convolutional Neural Network (CNN) for classification, a CNN for regression, and a Gradient Boosted Trees model using XGBoost for regression. These three models were tested on the S11 Amplitude and Phase features from the SansEC measurement and included all the frequency points and 391 data files collected during the 8-load cycle test conducted on September 10, 2024. The process began by labeling each data file (sample) with a corresponding strain value. The data was then grouped and labeled into distinct classes based on strain values. Each group of samples was sorted based on their strain value and assigned a class label. For example, Class 0 = 0.017804, Class 1 = 0.018199, and so on, reflecting ascending strain measurements. This mapping formed the basis of a structured dataset where each class is directly linked to a physical strain condition captured by the SansEC sensor.

### 8.1 Data Augmentation

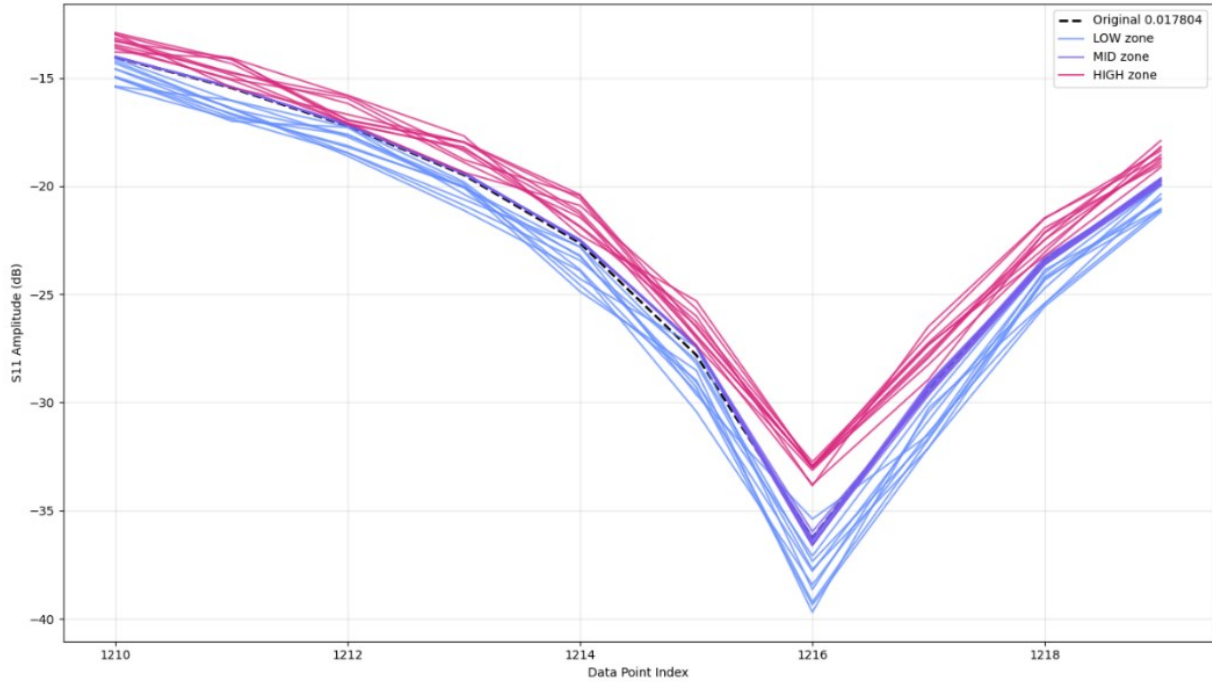
To increase the resolution and continuity of the dataset across the strain domain for improved model performance, synthetic data was generated using a triplet-based augmentation method using

S11 amplitude and phase features from the recorded SansEC measurement data files. Each data file was labeled based on its strain value. The augmentation process draws from three sequentially ordered strain classes referred to as low, middle, and high zones to form a localized window for interpolation. For each of these zones, new synthetic samples are generated by acquiring new values for S11 amplitude and phase. The demarcation for the low zone spans the interval below the low-to-middle midpoint, the middle zone lies between both midpoints, and the high zone extends beyond the middle to above the high midpoint. These bounds guide the creation of statistically perturbed feature values that preserve physical meaning while avoiding duplication with existing samples. Unlike arbitrary noise injection, this strategy introduces class-conditional variability similar in spirit to diffusion-based generative models yet anchored by real signal patterns associated with specific strain levels. The augmented samples maintain signal topology and are indistinguishable from real measurements in their structural behavior.

Figure 8.1.a presents a plot of the S11 amplitude versus data point index showing a comparison of the measurement data versus the synthetic data. The data point value correlates to the VNA frequency where zero corresponds to the start frequency of 20 MHz, and data point 1602 corresponds to the highest frequency of 180 MHz. Data point 1216 corresponds to the resonance at 141 MHz. This data was collected at 5000-pound load on the first upward cycle with a strain value of 0.017804. The solid lines are the synthetically generated augmentation data showing the low, middle and high zones. Figure 8.1.b presents the same plot showing a total of 10 samples of additional synthetic data for each zone.



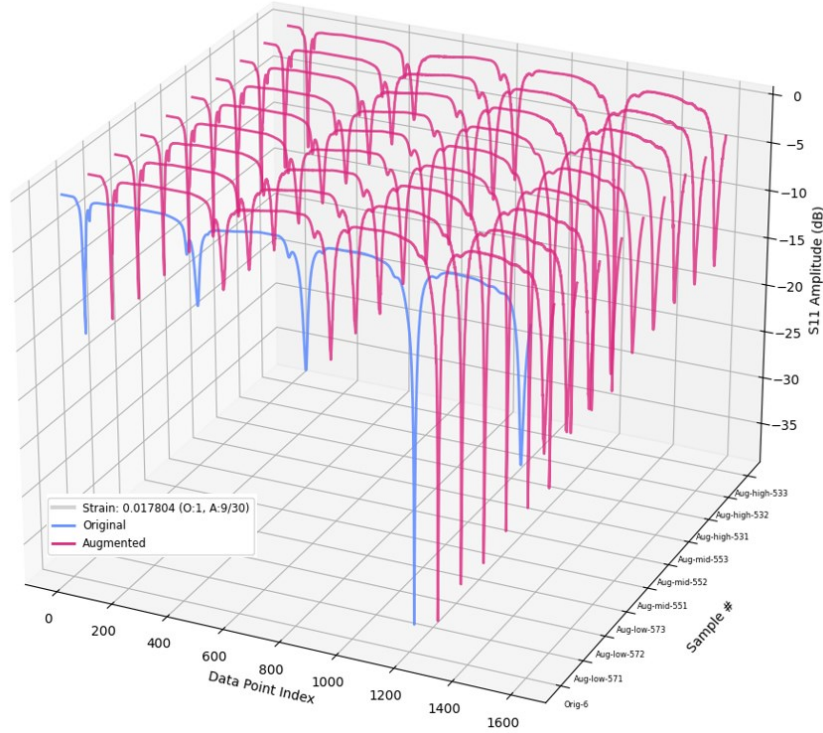
**Figure 8.1.a S11 Amplitude versus Data Point Index**



**Figure 8.1.b S11 Amplitude versus Data Point Index**

Figure 8.1.c presents a waterfall chart of the S11 amplitude versus data point index for the original measurement data (blue curve) and nine generated augmentation samples (pink curves), three from each triplet zone: low, middle, and high. Each curve represents the complete S11 frequency response of an individual sample. The augmented signals maintain the characteristic resonant behavior patterns while introducing controlled variability that preserves the signal's physical characteristics. This representation demonstrates how the triplet-based augmentation approach effectively generates realistic synthetic SansEC responses within each zone (low, middle, high) of the strain space, enabling more robust training of machine learning models without requiring additional physical measurements. The consistent pattern retention across all augmented samples validates that this technique produces physically plausible signals across the full spectrum of strain conditions.





**Figure 8.1.c Waterfall Chart of the S11 Amplitude versus Data Point Index**

## 8.2 Classification Training & Testing

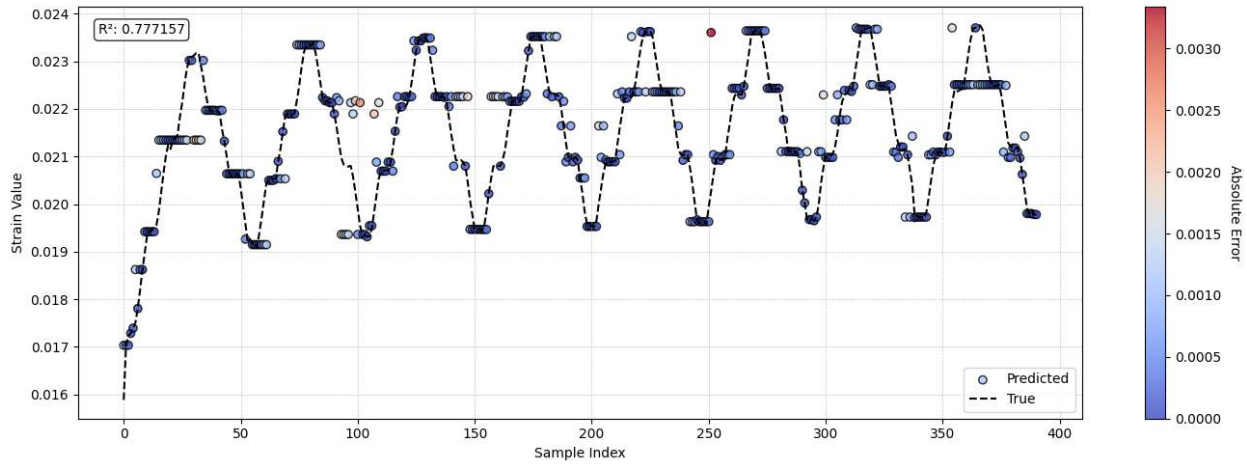
The three machine learning models were trained and validated on the S11 Amplitude and Phase from the synthetically generated samples. The purpose was to evaluate whether synthetic data alone, generated via triplet-based interpolation, could effectively train models that generalize well to real, unseen data. Training is defined as the process where the model learns patterns to develop potential functions by adjusting internal parameters that minimize error. Validation, in this context, means monitoring model performance on the synthetic dataset during training to ensure that learning is stable. After training, testing was done exclusively on the original unseen measurement data to evaluate its generalization to real-world measurements and quantify the  $R^2$  performance metric.

The CNN was trained to perform classification of strain levels based on SansEC responses. CNNs are a class of deep neural networks well-suited for structured data like time series and frequency-domain signals. They apply convolutional filters across the input to extract localized patterns enabling them to learn subtle variations in the signal that correspond to different physical conditions. In this experiment, the CNN model ingested a two-channel sequence of 1602 data points per sample: one channel for S11 Amplitude, and one for S11 Phase. Each sample was labeled with a specific strain value, and each unique strain value was treated as its own class label. This approach transformed the task into a fine-grained classification problem, where the model had to distinguish between small variations in signal tied to precise strain levels, rather than broad categories.

The model was trained and validated entirely on synthetic data samples, which were generated using the structured triplet-based interpolation technique to densely populate the strain space. The model was then tested on the original measurement samples, which were never seen during



training. The DIC strain measurement value corresponding to a measurement sample is defined as the “ground truth.” The CNN classifier achieved an  $R^2$  score of 0.777, indicating fair performance in recognizing and classifying real strain conditions based solely on high-resolution S11 signal inputs. Figure 8.2.a presents a comparison of the CNN model’s predicted ptrain versus the true strain measured from the DIC system for the 391 measurement samples. The y-axis strain values span the 376 unique strain classes in the measurement samples. The x-axis sample index spans the full 391 measurement samples tested. The black dashed line depicts the true strain over the eight load cycles comprised of an upward and downward load slope that is dynamically changing between 5000-pound and 8000-pound loads with momentary static hold load levels at every 1000-pound increment. The predicted strain values are depicted by the colored dots with a gradient scale from blue (lowest error) to red (highest error). The results show fairly good correlation while the strap is held at a constant load.



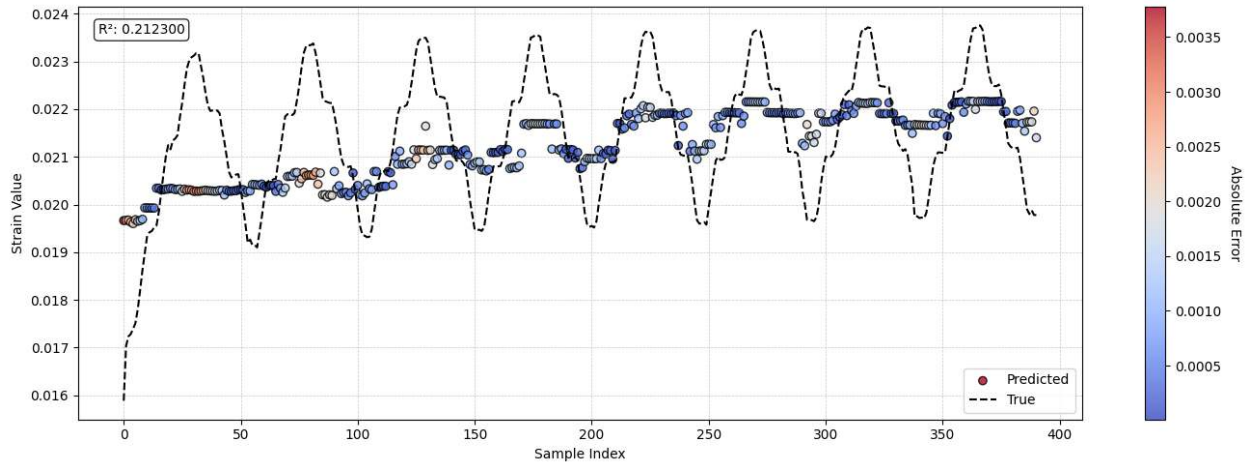
**Figure 8.2.a CNN Classification Model Predicted Strain versus True Strain**

### 8.3 Regression Training & Testing

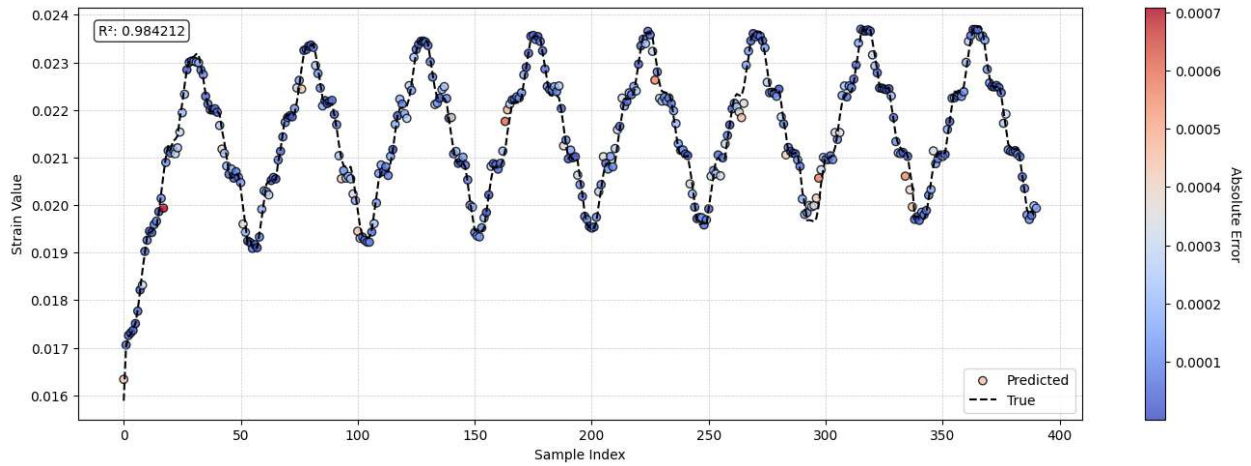
Two regression models were also developed and tested: a CNN built using TensorFlow, and a Gradient Boosted Decision Tree model built using XGBoost. Both models were trained and validated on the same synthetic dataset and tested exclusively on the original unseen measurement data to evaluate how well they generalize to quantifying strap strain. The models used the same S11 Amplitude and Phase, 1602 data points resulting in 3204 values per sample. These high-resolution signals captured the RF behavior of a Vectran strap under different strain levels and served as the input to both models. The CNN model was designed to process the sequential nature of the data, learning patterns across the signal using stacked layers of filters and transformation steps. In contrast, the XGBoost model treated the same input as a flat vector and applied decision tree-based logic to make predictions.

Figure 8.3.a presents the CNN Regression test results. The chart presentation is the same as described in Figure 8.2.a for the CNN classification model except the dots represent the prediction results from the CNN regression model. The CNN Regression model achieved an  $R^2$  score of 0.212. Figure 8.3.b presents the test results for XGBoost Regression model and follows the same chart descriptions as Figures 8.2.a and 8.3.a. The XGBoost model performed exceptionally well, achieving an  $R^2$  score of 0.984, showing strong alignment between predicted and actual strain

values. In summary, although both models were exposed to the same augmented data and tested on the same original measurement samples, the XGBoost regression model generalizes very well to unseen data, especially when using noise synthesized training data. The XGBoost model clearly outperformed both CNNs, highlighting the impact of model architecture on synthetic-to-real learning outcomes. A summary of the VOLT AI machine learning results is provided in Table 8.3.



**Figure 8.3.a CNN Regression Model Predicted Strain versus True Strain**



**Figure 8.3.b XGBoost Regression Model Predicted Strain versus True Strain**

**Table 8.3 Model Prediction Results from the CNN Classification, CNN Regression and XGBoost Regression Models Using All Eight Load Cycles.**

| <b>Machine Learning Method</b> | <b>Features</b>       | <b>Frequency Points</b> | <b>Training Samples</b> | <b>Testing Samples</b> | <b>Testing R<sup>2</sup></b> |
|--------------------------------|-----------------------|-------------------------|-------------------------|------------------------|------------------------------|
| CNN Classification             | S11 Amplitude & Phase | 1602                    | 11580                   | 391                    | .777                         |
| CNN Regression                 | S11 Amplitude & Phase | 1602                    | 11580                   | 391                    | .212                         |
| XGBoost Regression             | S11 Amplitude & Phase | 1602                    | 11580                   | 391                    | .984                         |

## 9.0 Sierra Turbines Machine Learning Summary

Sierra Turbines utilized supervised training of neural networks (NNs) as the core of their machine learning strategy. Neural networks were selected due to their universal function approximation capability, allowing them to capture the nonlinear, high-dimensional relationships inherent in the RF sensor data. NNs can model nonlinear, high-dimensional relationships commonly encountered in sensor response data. The models were built using a Python (open-source) neural training framework optimized by Optuna, an advanced hyperparameter tuning library that leverages Bayesian optimization (Tree-Structured Parzen Estimator) to efficiently explore optimal model configurations.

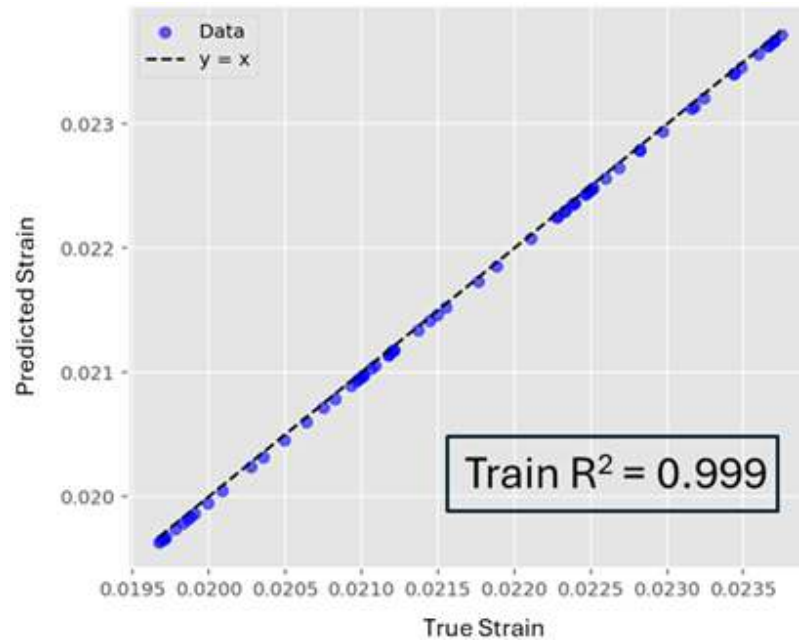
Optuna uses a history record of trials to determine which hyperparameter values to try next. NN hyperparameter values include, number of model layers, size of each layer, learning rate, dropout rate, alpha, and optimizer algorithm. Using this data, it estimates a promising area and does further evaluations in that area. Optuna then estimates an even more promising region based on the new result. It repeats this process using the history data of trials completed thus far, using a Bayesian optimization algorithm called Tree-Structured Parzen Estimator. Optuna supports asynchronous distributed optimization, which simultaneously performs multiple trials using multiple nodes. By making use of a combination of these functionalities, objective functions can be executed in a parallel, distributed manner. Optuna can conduct thousands of trials using many Bayesian optimizer algorithms (e.g., Adam, AdamW, Adamax, LBFGS, SGD, RMSprop, Rprop, Adadelta, etc.) to reduce the loss function efficiently.

SansEC data is well suited for machine learning since it can monitor the magnitude and phase at every frequency point within the sensor's operational bandwidth. What may appear to be a nonlinear observation at a single point in the analog domain may not be nonlinear in the machine learning domain. Deep learning models like neural networks are nonlinear by construction. They incorporate activation functions (rectifier, TanH, etc.) to produce nonlinear mappings. In contrast, analog circuit design treats nonlinearity at a single frequency as fatal, while deep learning treats it as undetectable noise unless it's part of a global, learnable trend. NN training should be effective

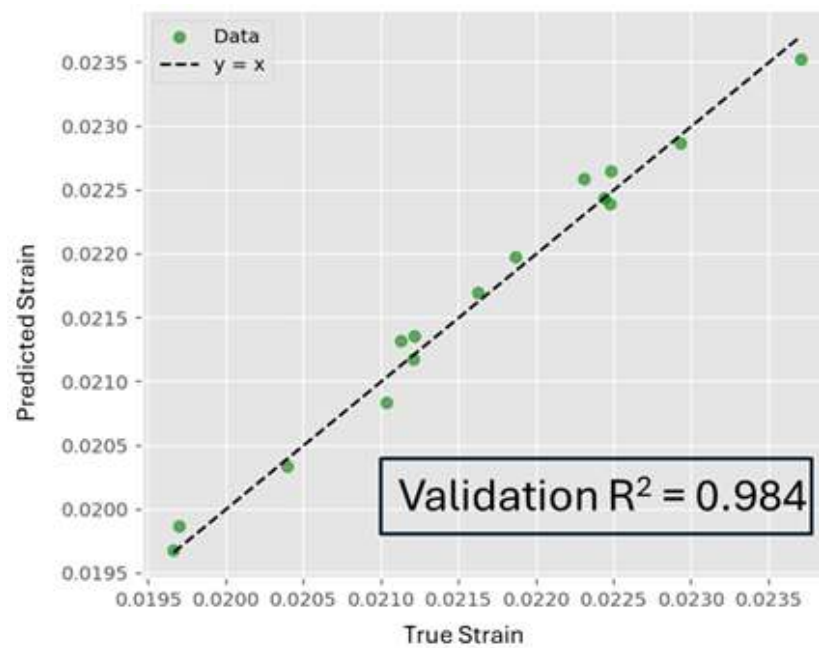
as long as the load-strain behaviour is repeatable. Since the goal of this experiment was to determine if the localized change in the dielectric values measured in the Vectran strap could be correlated to its strain using SansEC sensors, no assumptions about relative parametric importance or whether some portion of the swept spectrum was more critical than another. However, to reduce computation time, only three parameters (features) were modeled, S11 amplitude, S11 phase, and series reactance ( $X_s$ ).

### 9.1 Regression Training & Testing

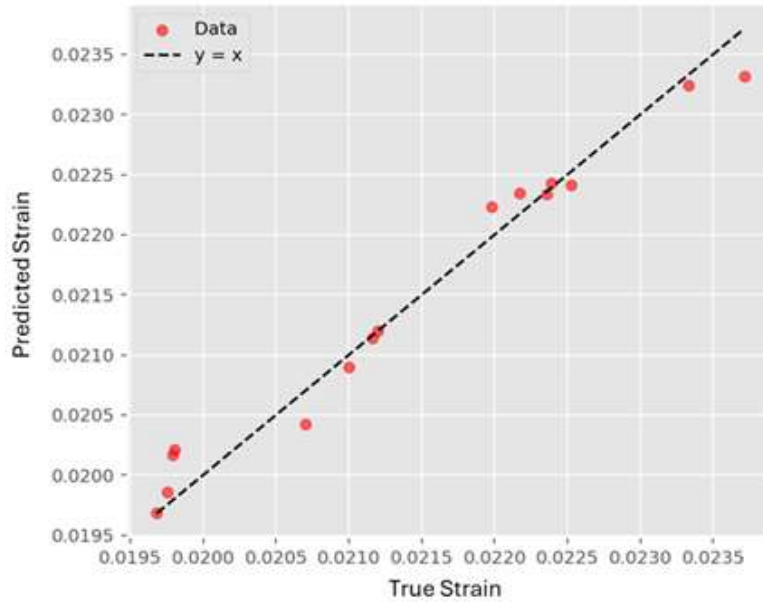
The last two load cycles (cycles seven and eight) from the NASA data collected on September 10, 2024 were ingested into a machine learning workflow. The workflow included typical normalization and scaling of features to facilitate faster and more reliable training. No Principal Component Analysis (PCA) was required to achieve high levels of  $R^2$ . A total of 97 measurements were used. The measurements were randomly divided into three groupings: 69 were used for training, 14 for validation, and 14 for test. Three features from the measurement data were used in the modeling: S11 amplitude, S11 phase, and the serial reactance ( $X_s$ , derived from amplitude and phase data using the impedance formula). All models used all 1602 frequency points. Figure 9.1.a presents the training results showing a comparison between the predicted strain (y-axis) versus the true strain (x-axis) for the 69 training measurements. The dashed black line represents zero error between the predicted and true strain values. The blue dots are the training predicted data points. The data points are sorted based on strain value from low to high. The model trained exceptionally well; it reproduced the true strain to within 0.2% error ( $R^2$  (test)  $\approx 0.999$ ). Figure 9.1.b presents the validation results showing a comparison between the predicted strain (y-axis) versus the true strain (x-axis) for the 14 validation measurements. The chart presentation is the same as Figure 9.1.a, except for the green dots, which represent validation prediction strain values. The model tracked the true values very closely ( $R^2 = 0.984$ ), though you can see a bit more deviation from the diagonal, indicating slightly increased error when generalizing beyond the training set. Figure 9.1.c presents the model test results showing a comparison between the predicted strain (y-axis) versus the true strain (x-axis) for the 14 test measurements. The chart presentation is the same as Figure 9.1.a and 9.1.b, except for the red dots which represent test prediction strain values. The model test performed very well ( $R^2 \approx 0.972$ ). This confirms robust generalization on unseen data. The Mean Absolute Error (MAE) was  $0.00011 \mu\epsilon$  with a root mean square error (RMSE) of  $2.0 \times 10^{-8} \mu\epsilon$ . A summary of the Sierra Turbines machine learning results is provided in Table 9.1



**Figure 9.1.a Predicted Training Model Strain versus True Strain**



**Figure 9.1.b Predicted Validation Model Strain versus True Strain**

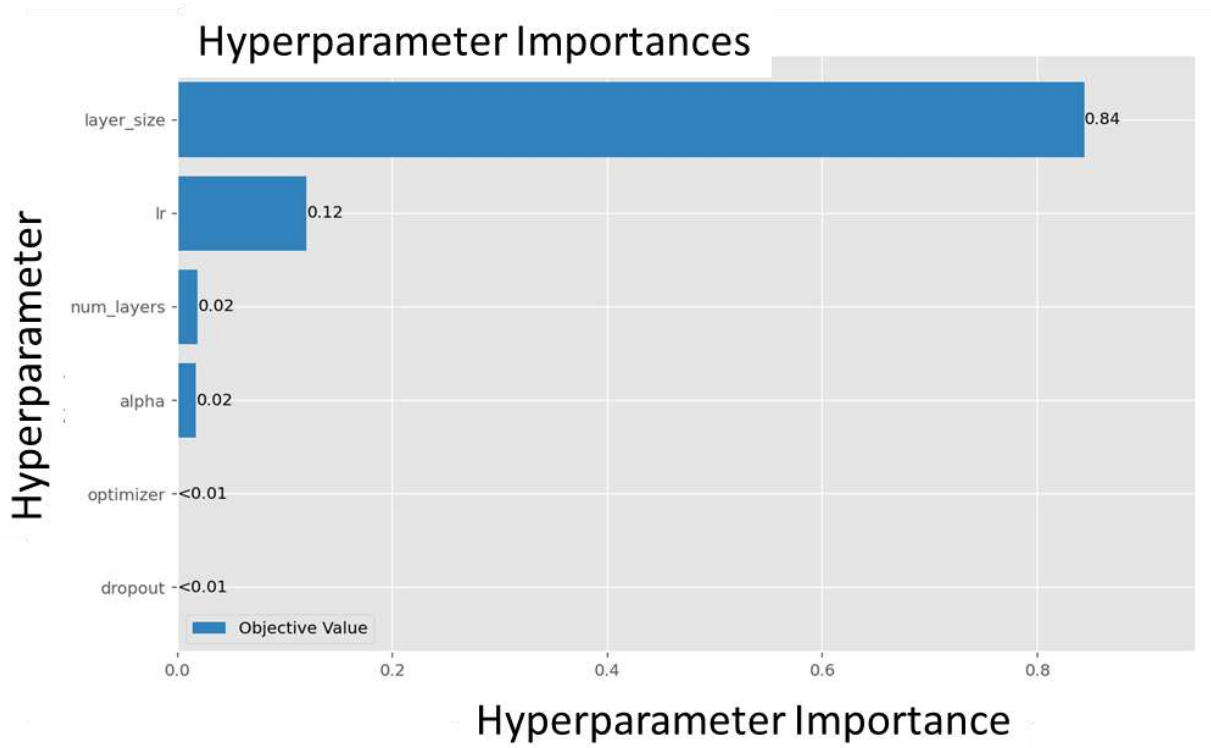


**Figure 9.1.c Predicted Test Model Strain versus True Strain**

**Table 9.1 Machine Learning Model Prediction Results from the NN Regression Model Using Only the Last Two of Eight Load Cycles.**

| Machine Learning Method                     | Features                      | Frequency Points | Training Samples | Validation Samples | Testing Samples | Testing $R^2$ |
|---------------------------------------------|-------------------------------|------------------|------------------|--------------------|-----------------|---------------|
| Supervised Neural Network Linear Regression | S11 Amplitude, S11 Phase & Xs | 1602             | 69               | 14                 | 14              | 0.975         |

Optuna identifies which neural network hyperparameters had the greatest impact on prediction accuracy [4]. As shown in Figure 9.1.d, the most influential factor was layer size, accounting for 84% of the variance in objective value. This suggests that the width of each layer strongly affects the model's capacity to learn from full-resolution SansEC signals. The learning rate (lr) contributed 12%, indicating its moderate role in convergence behavior. Other parameters such as the number of layers, L2 regularization strength (alpha), optimizer type, and dropout rate had minimal influence. This result justifies the use of wide, relatively shallow networks for modeling impedance-strain relationships from RF signals.



**Figure 9.1.d Neural Network Hyperparameter Importances**

## 9.2 Nonlinearity Simulation Results

To demonstrate neural network training immunity in the presence of nonlinearity, STI created a VNA simulation (both amplitude and phase) of a three-pole resonator as it is “tuned” across a parameter sweep. Each synthetic sweep consists of 1,000 frequency points covering 10 MHz to 70 MHz. At each frame, three Lorentzian-like resonances (originally centered at 20, 40, and 60 MHz) are collectively shifted by up to  $\pm 5$  MHz, then corrupted by Gaussian noise to meet a target SNR of 20 dB. The resulting magnitude (in decibels) and phase (in degrees) traces are stacked into feature vectors, saved as flat feature vectors for supervised NN training. We generated three VNA sweeps, one linear, two nonlinear (5% and 10% frequency reversal) with “bumps.” In the nonlinear runs, the bump occurs at midpoint of the sweep. By training these three scenarios in same manner as the SansEC, the results show that the 10% nonlinearity did not affect training accuracy. Detailed explanation, python code, and test results are available from Sierra Turbines upon request.

## 9.3 STI Modeling Observations

The results of the Sierra Turbines testing strongly support the feasibility of using SansEC sensors with advanced machine learning models for accurate strain detection. First, the neural network achieved a test  $R^2$  of approximately 0.972 using high-resolution S11 amplitude, phase, and derived serial reactance (Xs) data. This suggests that the model could explain 97.2% of the variance in the true strain measurements obtained by digital image correlation (DIC), demonstrating prediction accuracy comparable to traditional strain gauges. This provides strong evidence that SansEC sensors, when paired with nonlinear ML models, can deliver high-fidelity strain sensing. Also, note that only the real actual data was used with only typical scaling and normalization used (PyTorch StandardScaler).

Second, the use of the full 1602-point frequency-domain signal across multiple domains (amplitude, phase, and reactance) enabled the model to learn complex spectral-strain relationships without narrowing the data to specific resonance peaks. This indicates that the SansEC signal is information-rich and that ML algorithms are capable of extracting meaningful patterns for structural monitoring.

Third, the inclusion of derived physical features such as serial reactance improved the model's predictive performance and interpretability, pointing toward the value of incorporating physics-based features into ML pipelines. This further opens the door for future work on physics-informed neural networks.

Fourth, the testing focused on Cycles 7–8, which followed strap preconditioning, meaning the model primarily learned from stabilized, post-settlement behavior. The high performance in this regime suggests that SansEC + ML is well-suited for long-duration structural health monitoring in deployed habitat systems.

However, there are important caveats. Since the model was not trained on earlier load cycles, it does not capture dynamic phenomena such as hysteresis, early elasticity, or strap creep, which may be critical during inflation or first deployment phases. Therefore, the efficacy of SansEC strain estimation in these earlier mission phases remains uncertain. In addition, the model was only validated on a single strap under controlled conditions. There was no variation in strap material, temperature, or environmental drift, and sensor placement remained fixed. Thus, broader operational generalization, especially to multi-sensor deployments or embedded softgood systems still requires further testing.

Future work aimed at incorporating time history, transfer learning, and dimensionality reduction techniques is expected to enable highly accurate SansEC sensing even under sparse data regimes. This would allow models trained on limited or degraded signal inputs to retain performance, enhancing the practicality of SansEC-based strain monitoring in resource-constrained or dynamically evolving mission scenarios.

## 10.0 Strain Sensor Performance Requirements

In structural monitoring, small errors on the order of 5–10 microstrain ( $\mu\epsilon$ ) can be significant. A high  $R^2$  implies that the model explains at least 95% of the variance in the measured strain and that it aligns with expectations for replacing or augmenting a calibrated strain gauge. To gauge an MAE and RMSE requirement for this application, several factors need to be addressed in the future. For aerospace applications, where precision and reliability are paramount, there's no single universally acceptable MAE for strain gauges, as the acceptable MAE will vary based on a number of factors.

1. **Application criticality:** The acceptable MAE depends on the criticality of the component being monitored and the potential consequences of inaccurate strain measurements.
2. **Required accuracy:** The desired level of accuracy for the strain measurement, specified by design requirements.
3. **Environmental factors:** Temperature variations, humidity, vibration, and other environmental conditions in aerospace applications can significantly impact the performance and accuracy of strain gauges.



4. **Installation and bonding techniques:** Proper installation, adhesion, and bonding of the strain gauge are critical for accurate measurements and minimizing potential errors.
5. **Calibration and temperature compensation:** Calibration and appropriate temperature compensation techniques are essential to reduce MAE RMSE and ensure accurate readings, especially over wide temperature ranges [5]
6. **Strain gauges:** For less than 0.03% strain, strain gauges typically provide higher accuracy than DIC methods. It is possible that SansEC sensitivity and accuracy exceeds that of the photogrammetric laboratory reference [6].

## 11.0 Future Research

The work reported here was an initial effort to show the efficacy of using SansEC sensors to measure strain in a loaded Vectran strap. If future studies are conducted, there are several thoughts that should be considered. Multiple DIC measurements should be taken simultaneously along the length of the strap to determine the homogeneity of the axial strain along the length of the strap. Placing multiple SansEC sensors above and below the DIC measurement locations would provide a good assessment of the sensor's ability to measure localized strain. Long duration SansEC measurements on a strap held at a constant load would provide meaningful data to quantify creep strain. Taking the strap to failure would enable modeling of the full strap lifecycle. Conducting this test on multiple straps would establish generalization to the solution and confidence in the sensor's ability to provide structural health diagnostic and prognostic monitoring. More meaningful insights to monitor real time strap degradation might be possible with a faster VNA measurement. Limiting the frequency span to only include the most consequential resonance frequencies needed for an accurate prediction could achieve data rates down to a couple hertz or less. Optimizing the sensor geometry to match the desired data collection attributes is a logical next step. To explore strain inhomogeneities across the width of the strap as it ages under load, several small SansEC sensors could be placed across the width of the strap. This information could provide unique insights and support development of future strap compositions and weave designs. Improved robustness and performance in ML model designs should be expected from a continuation of this work.

## 12.0 Conclusions

SansEC sensor measurements were collected on a 12,500-pound rated Vectran strap to infer axial strain while it was loaded repeatedly between 5000- and 8000-pound loads. DIC measurements were simultaneously collected to provide true strain values to coincide with the SansEC response. After repeated cyclic loads, the strap strain was shown to steadily increase in value. However, the slope of the load strain curve remained constant. After repeated load tests on the same strap, measurement anomalies in the SansEC data indicated real time strap degradation had been observed.

A dielectric test set was used to measure the strap permittivity while under load. The measured change in the dielectric value was so small, it was considered to be within the uncertainty range of the measurement system. Absolute relative permittivity measurements were made on an unloaded 12,500-pound rated Vectran strap and the permittivity was in the range of 2.75 for frequencies from 200 MHz up to 20 GHz.

ResCon Technologies, VOLT AI, and Sierra Turbines were selected to participate in this NASA study and worked independently to develop machine learning models to predict strain from the SansEC measurement. Each partner established their own unique approach to modeling. ResCon was able to minimize the amount of data required to 40 frequency points and two data parameters, S11 amplitude and temporal index. VOLT AI demonstrated a novel technique to generate representative synthetic data to augment the training data set while maintaining signal topology. Sierra Turbines ingested the full 1602 frequency points using S11 amplitude, S11 phase, and series reactance (Xs) to develop the model. Each partner was able to successfully model the strap behavior and achieve an  $R^2$  above 0.97. This study has shown that SansEC sensors are able to measure strain on the Vectran strap and should be considered a candidate sensor to provide structural health diagnostic and prognostic monitoring for inflatable habitats.

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